NADI 2026:
The Second Multidialectal Arabic Speech Processing Shared Task

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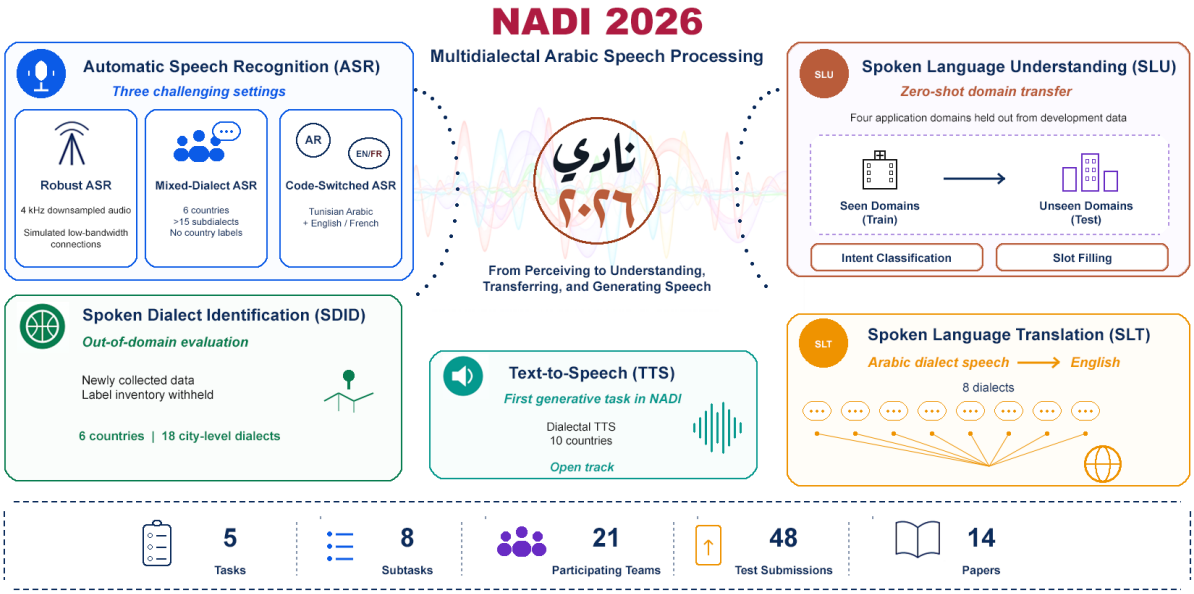


Figure 1: Overview of NADI 2026. The shared task spans the multidialectal Arabic spoken-language stack, covering speech perception through ASR and SDID, understanding through SLU, cross-language transfer through SLT, and speech generation through TTS. NADI 2026 further emphasizes realistic evaluation settings, including low-bandwidth, mixed-dialect, code-switched, out-of-domain, and zero-shot conditions. This edition comprises five tasks and eight subtasks, with 21 participating teams from at least 13 countries and 48 test-phase submissions.

Abstract

NADI 2026[1] is the seventh edition of the Nuanced Arabic Dialect Identification (NADI) shared task series and the second dedicated to multidialectal Arabic speech processing. This edition comprises five tasks and eight subtasks spanning Automatic Speech Recognition (ASR), Spoken Dialect Identification (SDID), Text-to-Speech (TTS), Spoken Language Translation (SLT), and Spoken Language Understanding (SLU). NADI 2026 emphasizes realistic evaluation through low-bandwidth, mixed-dialect, code-switched, out-of-domain, and zero-shot settings, while introducing TTS, SLT, and SLU to the series for the first time. The shared task attracted 21 participating teams from at least 13 countries, with 48 test-phase submissions, and 14 submitted system-description papers. Results show that out-of-domain generalization remains a bottleneck and highlight the effectiveness of recent Arabic-specialized speech models, multimodal dialect identification, and ensemble methods. Overall, NADI 2026 provides a broader and more challenging benchmark for robust Arabic dialect speech processing.

1 Introduction

Modern speech systems report steadily improving performance on Arabic benchmarks (Wang et al., 2025b),[2] yet speakers of Arabic dialects still meet systems that misidentify them, transcribe them

[1]NADI 2026 Official Website: https://nadi.dlnlp.ai/2026/

[2]Nearly a 10% absolute WER drop from 2024's (nvidia/conformer-ctc-large-ar) and the newly released audarai/Audar-ASR-V1-Turbo.

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poorly, or cannot speak to them in their own variety. Part of this gap is due to coverage: the International Organization for Standardization lists 28 active varieties under the ISO 639-3 Arabic macrolanguage code `ara`,[3] and even that figure understates city-to-city variation and the urban–rural divides. With few exceptions, benchmarks are usually assembled from clean, in-domain, and limited-variety recordings, whereas deployed systems encounter degraded audio, speakers whose variety is unknown in advance, utterances switching language or variety mid-sentence, and domains not anticipated at training time (Sullivan et al., 2026). A system can look stronger under the first set of conditions (characteristic of most benchmarks) than on the second.

Dialectal Arabic (DA) makes these challenges real. Unlike Modern Standard Arabic (MSA), the formal register for broadcast news and official addresses, DA is what speakers use in daily life and has no standard orthography (Ali et al., 2019, 2015). The same word is legitimately written in more than one way, making transcriptions inconsistent and *evaluation* itself less reliable. DA is also morphologically complex (Dahou et al., 2025), and code-switching into English, French, or MSA is far from exceptional (Dahou et al., 2025; Besdouri et al., 2024; Ali et al., 2021). Compounding this, some of the largest transcribed Arabic speech corpora, such as MGB2 (Ali et al., 2016) and QASR (Mubarak et al., 2021), are predominantly MSA. Dedicated dialectal corpora such as MASC (Al-Fetyani et al., 2022), ADI-20 (Elleuch et al., 2025), and Casablanca (Talafha et al., 2024) have begun to close this gap. Although the arrival of these dialectal corpora is precisely what makes evaluations reported here possible, they remain small compared to what MSA and high-resource languages enjoy.

The Nuanced Arabic Dialect Identification (NADI) shared task series was created to concentrate community efforts on this gap. It began in 2020 with country- and province level dialect identification over social media text (Abdul-Mageed et al., 2020), and across subsequent editions added variety identification across MSA and DA (Abdul-Mageed et al., 2021), dialectal sentiment analysis (Abdul-Mageed et al., 2022), dialect-to-MSA machine translation (Abdul-Mageed et al., 2023), and multi-label dialect identification with level-of-dialectness estimation (Abdul-Mageed et al., 2024). In 2025, NADI pivoted to speech, with a focus on spoken dialect identification, multidialectal automatic speech recognition (ASR), and spoken diacritic restoration (Talafha et al., 2025). Individual subtasks have come and gone, but dialect identification has run throughout the series: it is the tool that allows researchers to aggregate varieties, route utterances to dialect-specific models, and audit how well open resources cover the Arab world. Table 1 summarizes this trajectory and we provide a detailed comparison in Appendix A.1.

NADI 2026 is the seventh NADI shared task and the second dedicated to speech. Its design follows from the argument above and moves along two axes. The first is *realism*: rather than introducing new tasks, we re-introduce established ones under conditions that would break a system during deployment. ASR is evaluated on audio downsampled to 4 kHz and resampled to mimic low-bandwidth connections (Bouchakour et al., 2025), on mixed dialect speech (Besdouri et al., 2024) for which no country labels are supplied, and on Tunisian speech with frequent within-utterance switching into French or English. Spoken dialect identification (SDID) is evaluated out of domain (Sullivan et al., 2023; Abdullah et al., 2025), on newly collected data whose label inventory is withheld from participants, closing the loophole that lets system exploit a known label set (Sullivan et al., 2023; Abdullah et al., 2025).[4] Spoken language understanding reserves four application domains that appear nowhere the development data, thereby testing zero-shot domain transfer. The second axis is *scope*: with the addition of text-to-speech (TTS), spoken language translation (SLT), and spoken language understanding (SLU), NADI 2026 spans the full spoken-language stack for Arabic dialects. That is, perceiving speech (ASR, SDID), understanding it (SLU), transferring it across languages (SLT), and generating it (TTS). It is noteworthy that NADI 2026 introduces the first generative subtask in the history of the series, i.e., TTS.

NADI 2026 comprises five tasks and eight subtasks, all except TTS running on Codabench. Across a development phase running from 16 June to 20 July and a test phase from 20 to 28 July, we received 22 development-phase and 48 test-phase submissions, with 21 teams from at least 13 coun-

[3] https://iso639-3.sil.org/code/ara

[4] The inventory comprises six countries and 18 city-level dialects.

Comparison dimension	2020	2021	2022	2023	2024	2025	2026
Modality	Text	Text	Text	Text	Text	Speech	Speech
# Subtasks	2	4	2	3	3	3	8
Task families	T1	T1	T1, T2	T1, T3	T1, T3, T4	T1, T5, T6	T1, T5, T7–T9
Registered teams	61	53	41	58	51	44	65
Participating teams	18	8	21	18	12	8	21
Test submissions	56	68	105	76	76	100	48
Accepted papers	14	7	15	13	8	7	14

Table 1: Evolution of the NADI shared task series from 2020 to 2026. The series focused on text-based tasks from 2020 to 2024, transitioned to speech in 2025, and further expanded its speech-processing scope in 2026. Task-family codes are normalized across editions. **T1**: Dialect Identification; **T2**: Sentiment Analysis; **T3**: Dialect-to-MSA Machine Translation; **T4**: Arabic Level of Dialectness Estimation; **T5**: Automatic Speech Recognition; **T6**: Diacritic Restoration; **T7**: Text-to-Speech; **T8**: Spoken Language Translation; and **T9**: Spoken Language Understanding. For a detailed comparison of task coverage, task characteristics, geographic and variety scope, and evaluation metrics, see Appendix Table A.1.

tries submitting to at least one subtask. We received and accepted 14 system description papers.

A number of findings across the various shared task leaderboards are worth mentioning. First, out-of-domain generalization is a challenging problem: our dialect identification baseline falls from 81.19% accuracy in domain to 28.70% out of domain, and the best participating system recovers only to 56.15%. Second, no single recipe wins. For example, on code-switched ASR, a fully finetuned 2B parameter Arabic encoder–decoder, a LoRA adaptation training 3.6% of the parameters of its nearest rival, and a ten-system ROVER ensemble finish within 0.12 WER of one another. Third, Arabic-centric foundation models were utilized to great success this year, being used in five of the top six SDID entries, the top entry of one of the ASR subtasks, and second place entry in the other two ASR subtasks. This last finding is a visible shift from NADI 2025, which was dominated by general-purpose multilingual models. Fourth, system combinations helped almost everywhere.
We offer the following contributions:

- A *realism-first benchmark* covering recognition, identification, understanding, translation, and synthesis, each posed under a distribution shift rather than in-domain conditions.

- *New evaluation sets spanning five tasks*: previously unreleased portions of Casablanca (Talafha et al., 2024) including a low-bandwidth condition; a six-country, 18 city dialect identification set with withheld label inventory; dialectal TTS data covering ten countries; WhiteHouse (Bougares et al., 2026) evaluation data for dialect-to-English speech translation; and a new SLU evaluation set with four zero-shot domains.

- *A systematic analysis of the 21 participating systems*, 14 of which are documented in system description papers, identifying which adaptation, ensembling, and post-processing strategies generalize across subtasks and which do not.

The remainder of the paper is structured as follows: we provide a review of literature relevant to the shared task in §2; give an overview of this year's subtasks in §3; describe the baseline systems and team participation in §4; provide the results of each subtask in §5; discuss key findings in §6; and finally we conclude in §7.

2 Literature Review

Low-resource ASR increasingly uses transfer learning (Kopikar and Mandloi, 2026), self-supervised pretraining (Zhuo et al., 2025), and hybrid HMM/self-supervised architectures (Klejch et al., 2025). In-context learning offers another route for recognizing previously unseen languages (Li and Niehues, 2025), while large multilingual encoders have expanded coverage (Alabi et al., 2024). These methods are particularly relevant to Arabic dialects, which lack the transcribed resources available for MSA (Sullivan et al., 2026; Ali et al., 2017; Bouamor et al., 2018).

The NADI shared tasks have consequently promoted research on dialectal and code-switched Arabic (Talafha et al., 2025). Because standard-language ASR systems often perform poorly on

dialectal speech, recent Arabic research has investigated dialect-specific fine-tuning (Wang et al., 2025b; Ömer Tarik Özyilmaz et al., 2025) and joint dialect identification and speech recognition (Kumar et al., 2025). Spoken language identification has similarly progressed from i-vector (Dehak et al., 2011) and x-vector (Snyder et al., 2018) systems to Transformer-based models (Vaswani et al., 2017; Babu et al., 2022). Arabic dialect identification followed this transition (Ali et al., 2017; Shon et al., 2017; Ali et al., 2019; Elleuch et al., 2025; Talafha et al., 2025), although weak cross-domain generalization remains a persistent limitation (Sullivan et al., 2023; Abdullah et al., 2025).

Although DA speech processing has made strides in other areas, resource constraints pose a challenge for TTS (Laouirine et al., 2024) and SLT (Bougares et al., 2026; Abdulmumin et al., 2025; Bougares et al., 2025; Hamed et al., 2022) while SLU has mainly been focused on Tunisian Arabic (Mdhaffar et al., 2024a, 2025). We provide a more detailed review of related literature in Appendix A.

3 NADI 2026

This year we include five tasks, with additional subtasks for ASR and SLU. Except for TTS, all subtasks were evaluated on the Codabench platform, with a development phase and an evaluation phase. The remainder of this section gives a breakdown of each of the tasks/subtasks.

3.1 Task 1: ASR

Task Description. Our three ASR subtasks involve transcribing audio utterances from different dialectal, noisy, and domain conditions. The first of these subtasks, **1.1 Robust ASR**, emphasizes performance under a simulated low bandwidth scenario. The second, **1.2 Mixed Dialect ASR**, highlights the challenge of processing DA utterances without dialect labels. The final subtask, **1.3 Code-switched ASR** focuses on code-switching between Tunisian Arabic, English, and French.

Data. For the Robust ASR subtask, we apply a downsampling strategy to audio samples from the Casablanca (Talafha et al., 2024) dataset, with the test data coming from unreleased portions of the dataset. We downsample these files to 4khz and then resample them up to 16khz, in an effort to mimic extremely low bandwidth connections.

In the Mixed-Dialect subtask, we use newly collected data across six countries covering over 15 subdialects. No country labels are provided as part of the task.

Our Code-switched subtask targets transcribing spontaneous Tunisian Arabic speech with heavy French and English code-switching. Participants to this subtask are evaluated using newly collected data of around 1 hour and 50 minutes that represent 2,123 segments from a mix of talks and podcasts across diverse domains.

Evaluation Metrics. We use word error rate (WER) as our primary metric and character error rate (CER) as a secondary metric. For the official rankings we calculate these error rates without hamza normalization.

3.2 Task 2: Spoken Dialect Identification

Task Description. SDID involves estimating which of a closed set of dialects a spoken utterance belongs to. These tools have been observed to overfit to training domains (Sullivan et al., 2023; Abdullah et al., 2025), and thus this year we focus on out-of-domain SDID.

Data. Our target test data comes from newly collected data covering six countries and 18 city-level dialects within these six countries. Participants were not informed of the possible countries that would be included in the test data, except that they would be a subset of the 20 country labels used in ADI-20 (Elleuch et al., 2025).

Evaluation Metrics. We use two metrics for this task: Accuracy and the Average Cost metric from NIST Language Recognition Evaluation 2022 (Lee et al., 2022).

3.3 Task 3: Dialectal Text-to-Speech

Task Description. This task requires participants to synthesize dialectal Arabic speech from text prompts, capturing the characteristic pronunciation, rhythm, and prosody of the target dialect. Newly introduced to NADI this year, the task aims to establish benchmarking practices for an area where standard evaluation remains scarce. Given only text prompts, systems are evaluated on generating high-quality speech across ten Arab countries: Egypt, Sudan, Saudi Arabia, the United Arab Emirates, Jordan, Palestine, Tunisia, Algeria, Morocco, and Yemen. To encourage realistic acoustic quality and prosodic fidelity across diverse dialects, this task

was conducted as an open track, allowing participants to utilize any external training data.

Data. To build the evaluation dataset, we sampled sentences representing each target dialect from the Bulbul dataset (Ashraf et al., 2026).The text in Bulbul was manually collected and verified by native speakers from each respective country. Details of the data processing can be found in Appendix B.1.

Evaluation Metrics. To evaluate our task, we adopt two widely used neural non-intrusive metrics: UTMOSv2 (Baba et al., 2024) and NISQA (Mittag et al., 2021; Mittag and Möller, 2020), both of which leverage deep learning models trained on human perception ratings. To capture semantic performance, we also calculate WER and CER using the 7B parameter Omnilingual LLM-based ASR system with language set to 'arb_Arab' (Omnilingual ASR et al., 2025).

3.4 Task 4: Spoken Language Translation

Task Description. Participants in this task are expected to train systems that translate from eight Arabic dialect speech inputs into English. Systems must cope with dialectal variation, spontaneous speech, and real-world speaking conditions.

Data. Evaluation data is sourced from WhiteHouse (Bougares et al., 2026), the multi-dialectal Arabic speech translation dataset. WhiteHouse was created by supplementing the multi-dialectal Casablanca corpus with English translation. In total, the evaluation dataset comprises 6,788 and 6,780 utterances representing the validation and test sets of the eight Arabic dialects considered in the Casablanca dataset. Participants were provided with the WhiteHouse validation set for model development, while the test set, presented in Table B.2 was used for the official evaluation.

Evaluation Metrics. All participant submissions are evaluated with BLEU (Papineni et al., 2002) and chrF (Popović, 2015) scores. All submissions are evaluated with case-sensitivity and without punctuation. Scores are computed using the Sacrebleu toolkit (Post, 2018).

3.5 Task 5: Spoken Language Understanding

Task Description. SLU refers to natural language processing tasks that aim to extract semantic information from speech (Tur and De Mori, 2011). The goal of this task is to extract semantic information directly from Tunisian dialect speech, beyond literal transcription under spontaneous, possibly code-switched conditions. We break this into two subtasks **5.1 Intent Recognition** and **5.2 Slot Filling**. For intent classification, participants must classify spoken utterances into predefined intent categories capturing the speaker's goal. In the slot filling subtask, participants must identify and extract semantic information (entities, attributes, task-relevant details) from spoken utterances.

Data. Participants were provided with the original SLURP-TN (Elleuch et al., 2026a) training and development sets for system development. For evaluation, we introduce a new dataset recorded following the same guidelines and annotation protocol as SLURP-TN, ensuring consistency in terms of task design, intent and slot definitions, and data collection procedures. The new evaluation set contains 989 distinct utterances and amounts to one hour of clean speech. See Appendix B.3 for complete details.

Evaluation Metrics. The SLU slot filling task was evaluated based on terms of Concept Error Rate (CoER) and Concept/Value Error Rate (CVER). CoER is computed similarly to WER by taking into account only the semantic labels in the reference and hypothesis annotations. The CVER computation is identical, but the occurrences of concept/value pairs are taken into account instead of the concept alone. CVER implies that if any character within the word's support prediction or the concept tag prediction differs from the reference, the entire prediction for that concept is considered an error. CoER and CVER are detailed in (Laperrière et al., 2022).

4 Shared Task Teams & Baselines

4.1 Participating Teams

We provide an overview of participation across the phases in Table A.2, with a final total of 21 participating teams listed in Table 2.

4.2 Baselines

We use the `whisper-large-v3` model (Radford et al., 2023) as an off-the-shelf baseline for ASR and SLT tasks. For SDID we finetune an ECAPA-TDNN (Desplanques et al., 2020) model on a subset of the ADI20 dataset (Elleuch et al., 2025). For the TTS task, we have generated speech samples

Team Name	Affiliation	Subtask
A3da2 Messi	Global Brands Group (GBG)	5.1
Abjad AI (Alhumid et al., 2026)	Abjad AI, Jordan & Saudi Arabia	1.3, 2
AI Elites (Iskandar et al., 2026)	KAUST Academy, Saudi Arabia	2
Ar_India	Independent Researchers, India	1.1, 1.3, 2
ArabSpeech Minds	Nile University, Egypt	2
Aslema (Shafiq et al., 2026)	QCRI, Hamad Bin Khalifa University, Qatar	5.1, 5.2
CIS-OpenSLU (Khaled, 2026)	Nile University, Egypt	5.1
CodeZone (Yassine and Bichi, 2026)	Lebanese university, Lebanon & Vivekananda Global University, India	2
FARABI-Fraunhofer-IAIS	Fraunhofer IAIS, Germany	2, 5.1
Itgan (Almajai, 2026)	Independent Researcher, UK	1.1, 1.2, 1.3
Lynx (Atou et al., 2026)	Mohammed VI Polytechnic University, Morocco	2, 5.1
NAMAA (Abdelazi et al., 2026)	NAMAA Community, Saudi Arabia	1.2, 1.3, 2
Nile (Alshrief et al., 2026)	Nile University, Egypt	1.1
KAND CA (Wasfy and Anber, 2026)	KAND CA, Canada	1.1, 1.2, 1.3, 2, 3, 4
Resonate (Shoaib et al., 2026)	Nile University, Egypt	1.1, 2
Sabaa (Sabaa and Sabaa, 2026)	Najran University, Saudi Arabia	1.1
Salesteq	Salesteq, Switzerland	1.1, 1.2, 2
Sense	Birzeit University, Palestine	4
Thakaa (Otaif et al., 2026)	Thakaa, Advanced Company for Information Technology, Saudi Arabia	1.1, 1.3, 2
Wifaq (Alkurbi, 2026)	Independent Researcher, Saudi Arabia	1.1
Zila	Hacettepe University, Türkiye	5.2

Table 2: List of teams that registered and participated in NADI 2026 shared task. Teams with accepted papers are cited.

using XTTS-v2 [5] for all dialects. We choose only one reference speaker and there were no special prompting used. For SLU, we provide Whisper-based systems for both subtasks. The Intent Recognition baseline is a `whisper-small` encoder followed by an average pooling and classification layers. For Slot Filling, the full `whisper-small` system is used with its tokenizer expanded to include the semantic tags in the training data. For complete details of baseline models see § C.

5 Results

We summarize the results of the shared tasks as well as contrast the general strategies used. For descriptions of individual approaches, see § E.

5.1 Task 1: ASR

5.1.1 Subtask 1.1: Robust ASR

Table 3 summarizes the nine leaderboard entries. System descriptions were available for seven of the nine teams. Overall, the leaderboard does not identify a single winning recipe. WIFAQ (43.24% WER, 19.75% CER) led RESONATE by 7.05% absolute WER, whereas RESONATE (50.29% WER, 24.43% CER) and THAKAA (50.42% WER, 30.55% CER) were separated by only 0.13%, although with a much larger gap in CER.

[5] https://github.com/idiap/coqui-ai-TTS

The top three represented distinct strategies: WIFAQ combined four w2v-BERT 2.0 (Barrault et al., 2023b) CTC recognizers with five-gram KenLM (Heafield, 2011) decoding and ROVER (Fiscus, 1997); Resonate fully adapted one Arabic-specialized encoder-decoder model; and THAKAA used a heterogeneous, dialect-aware ROVER ensemble. Interestingly, while WIFAQ also dominated CER, KAND CA placed second on that metric, opting for a finetuned Whisper paired with a finetuned mT5 multidialect model (Xue et al., 2021) for spelling correction; the same strategy may explain its comparatively weaker WER, where it ranked fourth. Teams utilized diverse supplementary training data, with WIFAQ and KAND CA both including Casablanca (Talafha et al., 2024) and THAKAA using the Subtask 1.3 data, whereas the second-ranked team, RESONATE, avoided extra data completely. Dialect-aware mechanisms, such as dialect-specific routing, likewise appeared at both higher and lower ranks.

5.1.2 Subtask 1.2: Mixed-Dialect ASR

As shown in Table 4, SALESTEQ achieved the lowest WER at 44.22%, closely followed by NAMAA COMMUNITY at 44.65%. ITGAN and KAND CA obtained WERs of 46.71% and 48.07%, respectively. For CER, NAMAA COMMUNITY achieved the best score (17.16%), followed by KAND CA

Rank	Team	WER ↓	CER ↓	FT	LoRA	Add. data	Aug.	Dialect-aware	LM	Ensemble	Postproc.
1	Wifaq	**43.24**	**19.75**	✓		✓			✓	✓(4)	
2	Resonate	50.29	24.43	✓			✓	✓			✓
3	Thakaa	50.42	30.55	✓	✓	✓		✓	✓	✓(8)	
4	KAND CA	51.11	23.62	✓		✓		✓			✓
5	Ar_India	52.91	25.36								
6	Nile	56.90	27.67	✓		✓	✓				
7	Itgan	57.24	28.05		✓			✓			✓
8	Sabaa	59.77	28.04		✓			✓			✓
9	Salesteq	64.20	35.80								
10	Baseline	81.37	45.89								

Table 3: Official Subtask 1.1 (Robust ASR) results and reported system characteristics, ordered by country-average percentage WER and; percentage CER is also shown, and lower is better. **FT**: task-specific fine-tuning other than explicitly reported Low-Rank Adaptation (LoRA); **Add. data**: use of data beyond the official Subtask 1.1 data, excluding checkpoint pretraining; **Aug.**: acoustic augmentation; **Dialect-aware**: use of either separate adapters, conditioning, routing, or dialect-specific normalization; **LM**: external language model decoding; **Ensemble**: combination of multiple ASR systems (with number of components); **Postproc.**: denotes output transformations. A checkmark indicates a reported final-system component. AR_INDIA and SALESTEQ did not submit system papers.

		Results		System Features						
Rank	Team	WER ↓	CER ↓	Model	Adapt.	Add. Data	Ens.	Replay	VAD	Post.
1	**Salesteq**	**44.22**	18.05							
2	Namaa	44.65	**17.16**	Cohere	ZS					
3	Itgan	46.71	18.78	Whisper	LoRA	✓	✓(5)		✓	
4	KAND CA	48.07	17.28	Whisper	FFT	✓		✓		✓
5	Baseline	57.91	23.66	Whisper						

Table 4: Official results and main system features for Subtask 1.2 (Mixed-Dialect ASR) in percentage WER and CER. **Adapt.**: Model adaption strategy including zero-shot inference (ZS), low-rank adaptation (LoRA), and full-model fine-tuning (FFT); **Add. Data**: use of training data beyond the Task 1.2 development set; **Ens.**: combination of multiple ASR systems (and number of components); **Replay**: reuse of previously seen training examples during subsequent fine-tuning; **VAD**: voice activity detection; and **Post.**: explicit transcription post-processing. A ✓ indicates that the corresponding component was used. SALESTEQ did not submit a system paper.

(17.28%).

The submitted systems followed different strategies: NAMAA COMMUNITY used zero-shot inference with an Arabic-specific Cohere model (Cassini et al., 2026), ITGAN combined LoRA-adapted Whisper models using ROVER, while KAND CA relied on full-model fine-tuning with additional data and replay. The top ranked team, SALESTEQ, unfortunately did not submit a system paper. For data, most submissions did not use external data for this task, with only ITGAN using task 1.1 training data, and KAND CA using Casablanca (Talafha et al., 2024) to supplement training.

5.1.3 Subtask 1.3: Code-switched ASR

For code-switched ASR, the leaderboard, Table D.1, is exceptionally tight at the top: the first three entries span 0.12% absolute WER. Below that cluster, KAND CA trails the leader by 0.8% and NAMAA COMMUNITY by 2.57%, while AR_INDIA is a far outlier at 73.76%. Hamza normalization lowers the top five teams' WER by roughly 0.30% and does not reorder the ranking. As in Subtask 1.1, the two metrics give different views: ABJAD AI led on WER, whereas ITGAN recorded the lowest CER of the entire leaderboard, and THAKAA ranked third by WER but fourth by CER.

One strategy that clearly paid off in this subtask was the multi-*model* approach: all three top systems used some form of model combination (weight averaging or ROVER). As in Subtasks 1.1 and 1.2, supplementary training data approaches were diverse, with the ABJAD AI team utilizing synthetic speech, and the third-ranked team utilizing TUNIFRA (Choux et al., 2025) and Tunisian SLURP (Elleuch et al., 2026a).

5.2 Task 2: Spoken Dialect Identification

We provide the leaderboard for the SDID task in Table D.2. This task proved quite difficult compared to the in-domain validation performance,

with the baseline model going from 81.19% accuracy and C_{avg} of 10.60 to an accuracy of 28.70% and C_{avg} of 20.02. However, the teams were able to make a large gain over this performance, with THAKAA achieving the best accuracy at 56.15% and SALESTEQ achieving the best cost at 8.64.

Multimodal approaches appeared to dominate performance, with THAKAA using an ensemble mainly weighted towards a single acoustic model, Ara-BEST-RQ (Elleuch et al., 2026b), and lightly weighted (30%) text branches. A number of groups had reasonable success using Cohere's Arabic transcription model (Cassini et al., 2026), with AI ELITES, KAND CA, and ABJAD AI all using it as their base model for finetuning, while THAKAA used it as part of their text-pipelines.

5.3 Task 3: Text-to-Speech

This task received a single submission from KAND CA (Wasfy and Anber, 2026), covering all 4,001 evaluation samples across the ten target dialects. As shown in Table D.3, the system achieved aggregate scores of 2.72 UTMOS, 3.57 NISQA-MOS, and 3.21 NISQA-TTS, together with a WER of 32.48% and a CER of 11.40%. Performance varied across dialects: Morocco obtained the highest UTMOS score (3.23), while Sudan achieved the highest NISQA-MOS (4.46) and NISQA-TTS (3.98) scores. Saudi Arabia showed the strongest intelligibility results, with the lowest WER (14.28%) and CER (4.87%).

KAND CA used `lahgtna-omnivoice-v2`, a 612M-parameter OmniVoice-based model (Zhu et al., 2026) for multi-dialect Arabic speech synthesis. Zero-shot voice cloning was applied using reference speech selected for each target dialect, with generated audio produced at 24 kHz. Arabic diacritics were retained to support pronunciation during synthesis. For dialects without a directly supported reference configuration, the closest available dialect was used: Saudi Arabic for UAE and Palestinian Arabic for Jordan.

Comparison against the baseline is instructive. Averaged across dialects, the XTTS-v2 baseline outperforms the submission on all five metrics. This should be read with care. The baseline synthesizes all ten dialects from a single reference speaker with no dialect conditioning, and its perceptual scores are correspondingly near-invariant: UTMOS spans 0.06 across dialects and NISQA-MOS 0.18, against 1.06 and 1.81 for the submission. Metrics that barely register a system rendering every variety identically are largely insensitive to dialectal identity. UTMOSv2 and NISQA score general naturalness, while our WER and CER come from an ASR system carrying the same MSA bias this shared task exists to expose. Morocco is illustrative: the submission's highest UTMOS (3.23, above baseline) coincides with its highest WER (55.46% vs. 27.34%). We therefore take this result to reflect the limits of current automatic metrics rather than the futility of dialect-specific synthesis: informal listening confirms that the baseline renders every dialect in a largely uniform, MSA-leaning register that none of our metrics penalizes. Settling the matter properly requires human evaluation.

5.4 Task 4: Spoken Language Translation

As shown in Table D.4, SENSE achieved an overall BLEU score of 14.19 and a chrF score of 34.11. Dialect-level performance varied substantially, with the highest BLEU obtained for Jordan (27.46), followed by Palestine (20.32) and Egypt (19.84). Lower scores were observed for Morocco (8.51), Mauritania (6.84), and particularly the UAE (0.39).

Only the ABJAD AI team submitted a system paper. They directly finetuned Whisper Large-v3 on the provided training data, with additional data augmentation.

5.5 Task 5: Spoken Language Understanding

5.5.1 Subtask 5.1: Intent Recognition

As shown in Table D.5, LYNX achieved the best performance, with a weighted F_1 of 76.86% and an accuracy of 75.73%. CIS-OPENSLU ranked second with 71.54% weighted F_1 and 72.50% accuracy, followed by A3DA2 MESSI with 69.92% weighted F_1 and 70.07% accuracy, and then ASLEMA with 66.93% weighted F_1 and 66.13% accuracy. FARABI-FRAUNHOFER ranked fifth, obtaining 64.69% weighted F_1 and 65.93% accuracy. The ranking is consistent across both metrics, with LYNX maintaining a clear lead over the remaining systems.

The top two groups and the fourth-ranked team submitted system papers, with LYNX and CIS-OPENSLU both using an ASR-to-TF-IDF feature pipeline, while ASLEMA utilized a LoRA finetuned multimodal LLM instead. Neither LYNX nor CIS-OPENSLU utilized external data, while ASLEMA generated additional synthetic data for training. A novel feature of LYNX's approach was to filter unknown category data by training

MARBERTv2 (Abdul-Mageed et al., 2021) on MASSIVE (FitzGerald et al., 2023) and conditioning on the probability sum over the valid categories.

5.5.2 Subtask 5.2: Slot Filling

As shown in Table D.6, ASLEMA achieved the best performance, with a CoER of 59.53% and a CVER of 94.20%. ZILA ranked second with a CoER of 72.78% and a CVER of 100.36%.

Only ASLEMA provided a system paper, and their strategy was similar to their intent classification approach: LoRA finetuning `Qwen3-Omni-30B` (Yang et al., 2025a) on the SLURP-TN (Elleuch et al., 2026a) training split. Key to their strategy, however, was synthetic data generation, using `Gemini 3.6 Flash` and `3.1 Pro` to generate data, and `Gemini 3.6 Flash`, `3.1 Pro`, and `2.5 Pro` as a judging panel for data filtering.

6 Discussion

New model backbones. One model appeared frequently through different submissions: Cohere's `Transcribe` Arabic model (Cassini et al., 2026). It accounted for two of the top performing systems in Subtask 1.1, the second place system in Subtask 1.2, the top system in Subtask 1.3, four of the described systems in Subtask 2, and the top performing system in Subtask 5.1. Aside from this model, we observe that the **Ara-BEST-RQ** self-supervised learning model (Elleuch et al., 2026b) proved a powerful component for the top Subtask 2 team's submission.

Multimodal SDID. NADI 2025's SDID task was dominated by end-to-end acoustic models (Talafha et al., 2025), yet this year we have seen a number of multimodal systems perform well (accounting for three of the top five systems), hinting at the importance of both textual and spoken features. Historically, ASR transcripts (Malmasi and Zampieri, 2016) as well as phone-based features (Biadsy et al., 2009) have been utilized in SDID, and multimodal systems may reflect an evolution of these approaches.

Ensembling. Combined systems accounted for 10 of the 41 valid test submissions.[6] One real-world limitation of this approach, however, is the additional compute needed to both train and run inference of these models. To address low-compute scenarios, a potential direction for NADI would be to examine model efficiency across subtasks.

Is SDID Necessary? A surprising finding from the Mixed-Dialect ASR task, was the lack of any groups integrating SDID into their pipeline. The growth of multi-dialect models which do not differentiate between dialects well, such as Cohere's Transcribe Arabic or Whisper, may lead teams to embrace a 'one-size-fits all' approach to this task. Lack of data to train dialect-specific ASR models may contribute to this. Future work may benefit from directly comparing SDID-integrated pipelines and dialect-specific ASR with these more general approaches.

Lessons learned. Reflecting on this year's shared task, we have identified a few areas to consider before the next edition of NADI. First, the TTS evaluation should be redesigned (we expand on this further in §7). Second, the SDID subtask may be too difficult. It could benefit from providing dialect coverage information as well as being restructured to evaluate performance across several discrete domains, rather than as a purely zero-shot evaluation. Finally, an essential element of shared tasks is enabling teams to submit contrastive submissions. We believe we can achieve this by using evaluation platforms that better support display of variations in different systems.

7 Conclusion

For NADI 2026, the second speech processing focused edition of NADI, we have expanded into more challenging subtasks (Robust, Mixed-dialect, and Code-switched ASR) as well as introduced new subtasks (TTS, SLT, SLU). This year we also enjoyed the joint-largest number of participating teams (21, tied with NADI 2022), and the second-largest number of system description papers (14), while expanding to eight subtasks.

This year's shared task highlights a number of facets, including new pretrained Arabic ASR models with strong DA performance, the importance of multimodal approaches for SDID, and the dominance of ensemble-based approaches. In the future, we plan to examine performance across tasks under low-compute scenarios, more explicitly investigate the benefit of SDID systems as part of an ASR pipeline, and possibly expand the coverage to new dialects.

[6] We received 48 in total; see Table A.2 for the duplicate, unregistered, and invalid entries excluded

Limitations

Linguistic Coverage. While our objective is to provide coverage of a wide number of Arabic dialects in this shared task, we acknowledge the gap between our coverage and the great diversity of Arabic dialects across the world. In comparison to last year's NADI shared task, we believe we have made slight progress in this regard by introducing tasks that include code-switching and sub-dialectal variation. In terms of numbers of dialects, the broadest coverage task, TTS, still only covers ten dialects compared to the 28 varieties identified by ISO.

Dual Use Implications. While improving the spoken language processing methods for DA is essential in building equitable language technologies that match speaker's expectations for everyday language, these methods have potential for misuse, including for surveillance, or in the case of TTS, fraud or deception.

TTS Evaluation. We acknowledge two limitations with our TTS subtask: low participation and lack of human evaluation. With respect to low participation, one team communicated an expectation that training data would be provided and withdrew on hearing this was not the case. Additionally, while other subtasks may have benefited from CodaBench's discoverability features, information about the TTS subtask was only disseminated through our website and emails. Likewise, lack of a submission platform may have inconvenienced participants, further discouraging participation.

Next, while automatic, non-intrusive, metrics may capture some audio quality information and even some semantic information (via ASR WER), however, they lack a ground truth human evaluation to validate their performance and suitability across different Arabic dialects. While this year we focused on covering a wide number of dialects, an alternative would have been to provide lesser coverage of dialects, but in turn supplement the evaluation with human metrics.

These two factors limit the conclusions that can be drawn from this subtask, but does provides important organizational lessons which we hope will help address these issues in future iterations of NADI.

Orthographic Variability. Dialectal Arabic has no standardized orthography, yet WER and CER score a hypothesis against a single reference, so a transcription a native speaker would accept can be counted as wrong for choosing a different valid spelling. Subtask 1.3 lets us bound this: collapsing hamza variants alone lowers the top five teams' WER by 0.29 points, about 2% of measured word error, without reordering the leaderboard. Rankings therefore appear robust, but absolute rates are less so: hamza is one axis among several, so 2% is a lower bound. Multi-reference WER (Ali et al., 2015) was proposed for this setting and could be adopted in future editions.

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References

Mohamed Abdelazi, Fatimah Mohamed Emad Elden, Mousa Ashraf Abd El Malak, and Omer Nacar. 2026. NAMAA Community at NADI 2026: Mixed-Dialect and Code-Switched Arabic ASR and Spoken Dialect Identification. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Muhammad Abdul-Mageed, AbdelRahim Elmadany, and El Moatez Billah Nagoudi. 2021. ARBERT & MARBERT: Deep bidirectional transformers for Arabic. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7088–7105, Online. Association for Computational Linguistics.

Muhammad Abdul-Mageed, AbdelRahim Elmadany, Chiyu Zhang, El Moatez Billah Nagoudi, Houda Bouamor, and Nizar Habash. 2023. NADI 2023: The fourth nuanced Arabic dialect identification shared task. In *Proceedings of ArabicNLP 2023*, pages 600–613, Singapore (Hybrid). Association for Computational Linguistics.

[7] https://alliancecan.ca

Muhammad Abdul-Mageed, Amr Keleg, AbdelRahim Elmadany, Chiyu Zhang, Injy Hamed, Walid Magdy, Houda Bouamor, and Nizar Habash. 2024. NADI 2024: The fifth nuanced Arabic dialect identification shared task. In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 709–728, Bangkok, Thailand. Association for Computational Linguistics.

Muhammad Abdul-Mageed, Chiyu Zhang, Houda Bouamor, and Nizar Habash. 2020. NADI 2020: The first nuanced Arabic dialect identification shared task. In *Proceedings of the Fifth Arabic Natural Language Processing Workshop, WANLP@COLING 2020, Barcelona, Spain (Online), December 12, 2020*, pages 97–110. Association for Computational Linguistics.

Muhammad Abdul-Mageed, Chiyu Zhang, AbdelRahim Elmadany, Houda Bouamor, and Nizar Habash. 2022. NADI 2022: The third nuanced Arabic dialect identification shared task. In *Proceedings of the Seventh Arabic Natural Language Processing Workshop (WANLP)*, pages 85–97, Abu Dhabi, United Arab Emirates (Hybrid). Association for Computational Linguistics.

Muhammad Abdul-Mageed, Chiyu Zhang, Abdel-Rahim A. Elmadany, Houda Bouamor, and Nizar Habash. 2021. NADI 2021: The second nuanced Arabic dialect identification shared task. In *Proceedings of the Sixth Arabic Natural Language Processing Workshop, WANLP 2021, Kyiv, Ukraine (Virtual), April 9, 2021*, pages 244–259. Association for Computational Linguistics.

Badr M. Abdullah, Matthew Baas, Bernd Möbius, and Dietrich Klakow. 2025. Voice Conversion Improves Cross-Domain Robustness for Spoken Arabic Dialect Identification. In *Interspeech 2025*, pages 2790–2794.

Idris Abdulmumin, Victor Agostinelli, Tanel Alumäe, Antonios Anastasopoulos, Luisa Bentivogli, Ondřej Bojar, Claudia Borg, Fethi Bougares, Roldano Cattoni, Mauro Cettolo, Lizhong Chen, William Chen, Raj Dabre, Yannick Estève, Marcello Federico, Mark Fishel, Marco Gaido, Dávid Javorský, Marek Kasztelnik, and 33 others. 2025. Findings of the IWSLT 2025 evaluation campaign. In *Proceedings of the 22nd International Conference on Spoken Language Translation (IWSLT 2025)*, pages 412–481, Vienna, Austria (in-person and online). Association for Computational Linguistics.

Mohammad Al-Fetyani, Muhammad Al-Barham, Gheith A. Abandah, Adham Alsharkawi, and Maha Dawas. 2022. MASC: Massive Arabic speech corpus. In *Proceedings of the 2022 IEEE Spoken Language Technology Workshop (SLT)*, page 1002206.

Jesujoba O Alabi, Xuechen Liu, Dietrich Klakow, and Junichi Yamagishi. 2024. AfriHuBERT: A self-supervised speech representation model for African languages. *arXiv preprint arXiv:2409.20201*.

Hatim Alhumid, Naif Alharthi, Elaf S. Alsaedi, Faris Alasmary, Ahmad Ghannam, Anas Adel Salamah, Shouq Sadah, and Lahouari Ghouti. 2026. Abjad AI at NADI 2026 shared task: Overcoming Data Scarcity in Arabic Speech Processing through Augmentation. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Ahmed Ali, Peter Bell, James Glass, Yacine Messaoui, Hamdy Mubarak, Steve Renals, and Yifan Zhang. 2016. The MGB-2 challenge: Arabic multi-dialect broadcast media recognition. In *2016 IEEE Spoken Language Technology Workshop (SLT)*, pages 279–284. IEEE.

Ahmed Ali, Shammur Chowdhury, Mohamed Afify, Wassim El-Hajj, Hazem Hajj, Mourad Abbas, Amir Hussein, Nada Ghneim, Mohammad Abushariah, and Assal Alqudah. 2021. Connecting Arabs: Bridging the gap in dialectal speech recognition. *Communications of the ACM*, 64(4):124–129.

Ahmed Ali, Walid Magdy, Peter Bell, and Steve Renais. 2015. Multi-reference WER for evaluating ASR for languages with no orthographic rules. In *2015 IEEE Workshop on Automatic Speech Recognition and Understanding (ASRU)*, pages 576–580. IEEE.

Ahmed Ali, Suwon Shon, Younes Samih, Hamdy Mubarak, Ahmed Abdelali, James Glass, Steve Renals, and Khalid Choukri. 2019. The MGB-5 challenge: Recognition and dialect identification of dialectal Arabic speech. In *2019 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, pages 1026–1033. IEEE.

Ahmed Ali, Stephan Vogel, and Steve Renals. 2017. Speech recognition challenge in the wild: Arabic MGB-3. In *2017 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, pages 316–322. IEEE.

Sulaiman Alkurbi. 2026. Wifaq at NADI 2026 Shared Task: Diversity-Aware System Combination for Robust Arabic ASR. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Ibrahim Almajai. 2026. Itgan at NADI 2026 shared task: Parameter-Efficient Whisper Adaptation for Robust, Mixed-Dialect and Code-Switched Arabic ASR. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Mahmoud Alshrief, Mohamed Bahgat, Mahmoud Zaher, and Sahar Selim. 2026. A Two-Phase Acoustic Fine-Tuning Strategy for Robust Dialectal Arabic Speech Recognition NADI 2026 Subtask 1.1: Noisy Country-Level ASR. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Junyi Ao, Rui Wang, Long Zhou, Chengyi Wang, Shuo Ren, Yu Wu, Shujie Liu, Tom Ko, Qing Li, Yu Zhang, Zhihua Wei, Yao Qian, Jinyu Li, and Furu Wei. 2022. SpeechT5: Unified-modal encoder-decoder pre-training for spoken language processing. *Preprint*, arXiv:2110.07205.

Rosana Ardila, Megan Branson, Kelly Davis, Michael Kohler, Josh Meyer, Michael Henretty, Reuben Morais, Lindsay Saunders, Francis Tyers, and Gregor Weber. 2020. Common Voice: A massively-multilingual speech corpus. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 4218–4222, Marseille, France. European Language Resources Association.

Ahmed Ashraf, Aisha Alansari, Fadel Al Abbas, Nada Almarwani, Samah Aloufi, Saad Ezzini, Maged S. Al-shaibani, Doaa Dalaq, AbdelRahim A. Elmadany, Muhammad Abdul-Mageed, Mohamed Mehdi Trigui, Dania Refai, Layan Refai, Mohamed Akrout, Mustafa Jarrar, Wasfi G. Al-Khatib, Alaa Dalaq, Darin El-Nakla, Samir Abdaljalil, and 14 others. 2026. Bulbul: A dataset for dialectal Arabic speech recognition. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, Budapest, Hungary. Association for Computational Linguistics. To appear.

Atlasia. 2026. MoulSot: A curated Moroccan Darija speech dataset and fine-tuned ASR model.

Houdaifa Atou, Issam Ait Yahia, and Ismail Berrada. 2026. Lynx at the NADI 2026 Shared Task: Exploring ASR Representations and Transcripts for Dialectal Arabic Speech Understanding. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Matthew Baas, Benjamin van Niekerk, and Herman Kamper. 2023. Voice conversion with just nearest neighbors. In *Interspeech 2023*, pages 2053–2057.

Kaito Baba, Wataru Nakata, Yuki Saito, and Hiroshi Saruwatari. 2024. The T05 system for the Voice-MOS Challenge 2024: Transfer learning from deep image classifier to naturalness MOS prediction of high-quality synthetic speech. In *IEEE Spoken Language Technology Workshop (SLT)*, pages 818–824.

Arun Babu, Changhan Wang, Andros Tjandra, Kushal Lakhotia, Qiantong Xu, Naman Goyal, Kritika Singh, Patrick von Platen, Yatharth Saraf, Juan Pino, Alexei Baevski, Alexis Conneau, and Michael Auli. 2022. XLS-R: Self-supervised Cross-lingual Speech Representation Learning at Scale. In *Interspeech 2022*, pages 2278–2282.

Alexei Baevski, Arun Babu, Wei-Ning Hsu, and Michael Auli. 2023. Efficient self-supervised learning with contextualized target representations for vision, speech and language. In *International conference on machine learning*, pages 1416–1429. PMLR.

Alexei Baevski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. 2020. wav2vec 2.0: A framework for self-supervised learning of speech representations. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 12449–12460.

Raphaël Bagat, Irina Illina, and Emmanuel Vincent. 2025. Mixture of LoRA Experts for Low-Resourced Multi-Accent Automatic Speech Recognition. In *Interspeech 2025*, pages 1143–1147.

Loïc Barrault, Yu-An Chung, Mariano Cora Meglioli, David Dale, Ning Dong, Paul-Ambroise Duquenne, Hady Elsahar, Hongyu Gong, Kevin Heffernan, John Hoffman, and 1 others. 2023a. SeamlessM4T: Massively multilingual & multimodal machine translation. *arXiv preprint arXiv:2308.11596*.

Loïc Barrault, Yu-An Chung, Mariano Coria Meglioli, David Dale, Ning Dong, Mark Duppenthaler, Paul-Ambroise Duquenne, Brian Ellis, Hady Elsahar, Justin Haaheim, and 1 others. 2023b. Seamless: Multilingual expressive and streaming speech translation. *arXiv preprint arXiv:2312.05187*.

Fatma Zahra Besdouri, Inès Zribi, and Lamia Hadrich Belguith. 2024. Arabic automatic speech recognition: challenges and progress. *Speech Communication*, 163:103110.

Fadi Biadsy, Julia Hirschberg, and Nizar Habash. 2009. Spoken Arabic dialect identification using phonotactic modeling. In *Proceedings of the EACL 2009 workshop on computational approaches to semitic languages*, pages 53–61.

Astik Biswas, Oleg Shevelev, Amine Abdaoui, Vivek Tyagi, and Abdelmoumene Boumadane. 2025. Adapting Whisper for low-resource Hindi-English code-mix speech with on-the-fly augmentation & LLM-synthesised data. In *Interspeech 2025*, pages 4293–4297.

Verena Blaschke, Miriam Winkler, Constantin Förster, Gabriele Wenger-Glemser, and Barbara Plank. 2025. A Multi-Dialectal Dataset for German Dialect ASR and Dialect-to-Standard Speech Translation. In *Interspeech 2025*, pages 913–917.

Houda Bouamor, Nizar Habash, Mohammad Salameh, Wajdi Zaghouani, Owen Rambow, Dana Abdulrahim, Ossama Obeid, Salam Khalifa, Fadhl Eryani, Alexander Erdmann, and Kemal Oflazer. 2018. The MADAR Arabic dialect corpus and lexicon. In *Proceedings of the Eleventh International Conference on Language Resources and Evaluation, LREC 2018, Miyazaki, Japan, May 7-12, 2018*. European Language Resources Association (ELRA).

Lallouani Bouchakour, Khaled Lounnas, and Ahmed Krobba. 2025. Enhancing Arabic ASR in noisy and transcoding EVS conditions: A multimodal deep learning study. In *FedCSIS (Position Papers)*, pages 1–8.

Fethi Bougares, Salima Mdhaffar, Haroun Elleuch, and Yannick Estève. 2025. TEDxTN: A three-way speech translation corpus for code-switched Tunisian Arabic - English. In *Proceedings of The Third Arabic Natural Language Processing Conference*, pages 278–287, Suzhou, China. Association for Computational Linguistics.

Fethi Bougares, Salima Mdhaffar, and Yannick Estève. 2026. WhiteHouse: Translation of the Casablanca corpus for multi-dialectal Arabic speech translation. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference*, pages 5849–5855, Palma de Mallorca, Spain. ELRA Language Resource Association.

Edresson Casanova, Kelly Davis, Eren Gölge, Görkem Göknar, Iulian Gulea, Logan Hart, Aya Aljafari, Joshua Meyer, Reuben Morais, Samuel Olayemi, and Julian Weber. 2024. XTTS: a massively multilingual zero-shot text-to-speech model. *Preprint*, arXiv:2406.04904.

Shaun Cassini, Sebastian Vincent, Xiaolu Lu, Julian Mack, Dhruti Joshi, and Pierre Richemond. 2026. cohere-transcribe-arabic-07-2026 (revision 0a8193c).

William Chan, Navdeep Jaitly, Quoc Le, and Oriol Vinyals. 2016. Listen, attend and spell: A neural network for large vocabulary conversational speech recognition. In *Proceedings of ICASSP*, pages 4960–4964.

Chaimae Chellaf, Haroun Elleuch, Othman Istaiteh, D. Fortuné Kponou, Fethi Bougares, Yannick Estève, and Salima Mdhaffar. 2025. LIA and ELYADATA systems for the IWSLT 2025 low-resource speech translation shared task. In *Proceedings of the 22nd International Conference on Spoken Language Translation (IWSLT 2025)*, pages 274–281, Vienna, Austria (in-person and online). Association for Computational Linguistics.

Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, and 1 others. 2022. WavLM: Large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing*, 16(6):1505–1518.

Yushen Chen, Junzhe Liu, Yujie Tu, Zhikang Niu, Yuzhe Liang, Chunyu Qiang, Chen Zhang, Kai Yu, and Xie Chen. 2026. Habibi: Laying the open-source foundation of unified-dialectal Arabic speech synthesis. *Preprint*, arXiv:2601.13802.

Alex Choux, Marko Avila, Josep Crego, Fethi Bougares, and Antoine Laurent. 2025. TuniFra: A Tunisian Arabic speech corpus with orthographic transcriptions and French translations. In *Proceedings of The Third Arabic Natural Language Processing Conference*, pages 64–68, Suzhou, China. Association for Computational Linguistics.

Alexis Conneau, Min Ma, Simran Khanuja, Yu Zhang, Vera Axelrod, Siddharth Dalmia, Jason Riesa, Clara Rivera, and Ankur Bapna. 2023. FLEURS: Few-shot learning evaluation of universal representations of speech. In *2022 IEEE Spoken Language Technology Workshop (SLT)*, pages 798–805. IEEE.

Abdelghani Dahou, Abdelhalim Hafedh Dahou, Mohamed Amine Chéragui, Amin Abdedaiem, Mohammed AA Al-Qaness, Mohamed Abd Elaziz, Ahmed A Ewees, and Zhonglong Zheng. 2025. A survey on dialect Arabic processing and analysis: Recent advances and future trends. *ACM Transactions on Asian and Low-Resource Language Information Processing*, 24(8):1–45.

Najim Dehak, Pedro A Torres-Carrasquillo, Douglas Reynolds, and Reda Dehak. 2011. Language recognition via i-vectors and dimensionality reduction. In *Interspeech 2011*, pages 857–860.

Brecht Desplanques, Jenthe Thienpondt, and Kris Demuynck. 2020. ECAPA-TDNN: Emphasized Channel Attention, Propagation and Aggregation in TDNN Based Speaker Verification. In *Interspeech 2020*, pages 3830–3834.

Spandan Dey, Md Sahidullah, and Goutam Saha. 2023. Cross-corpora spoken language identification with domain diversification and generalization. *Computer Speech & Language*, 81:101489.

ElevenLabs. 2023. Eleven Multilingual v2. `https://elevenlabs.io/blog/eleven-multilingual-v2`. Accessed: Aug. 9, 2026.

Haroun Elleuch, Salima Mdhaffar, Yannick Estève, and Fethi Bougares. 2026a. SLURP-TN : Resource for Tunisian dialect spoken language understanding. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference*, pages 1544–1552, Palma de Mallorca, Spain. ELRA Language Resource Association.

Haroun Elleuch, Salima Mdhaffar, Yannick Estève, and Fethi Bougares. 2025. ADI-20: Arabic Dialect Identification dataset and models. In *Interspeech 2025*, pages 2775–2779.

Haroun Elleuch, Youssef Saidi, Salima Mdhaffar, Yannick Estève, and Fethi Bougares. 2025. ELYADATA & LIA at NADI 2025: ASR and ADI subtasks. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 762–766, Suzhou, China. Association for Computational Linguistics.

Haroun Elleuch, Ryan Whetten, Salima Mdhaffar, Yannick Estève, and Fethi Bougares. 2026b. Ara-BEST-RQ: Multi dialectal Arabic SSL. In *ICASSP 2026-2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 5326–5330. IEEE.

Jonathan G Fiscus. 1997. A post-processing system to yield reduced word error rates: Recognizer output voting error reduction (ROVER). In *1997 IEEE workshop on automatic speech recognition and understanding proceedings*, pages 347–354. IEEE.

Jack FitzGerald, Christopher Hench, Charith Peris, Scott Mackie, Kay Rottmann, Ana Sanchez, Aaron Nash, Liam Urbach, Vishesh Kakarala, Richa Singh, Swetha Ranganath, Laurie Crist, Misha Britan, Wouter Leeuwis, Gokhan Tur, and Prem Natarajan. 2023. MASSIVE: A 1M-example multilingual natural language understanding dataset with 51 typologically-diverse languages. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4277–4302, Toronto, Canada. Association for Computational Linguistics.

Xun Gong, Anqi Lv, Wangyou Zhang, Zhiming Wang, Huijia Zhu, and Yanmin Qian. 2025. BR-ASR: Efficient and Scalable Bias Retrieval Framework for Contextual Biasing ASR in Speech LLM. In *Interspeech 2025*, pages 4043–4047.

Google Cloud. 2026. Gemini-TTS | Cloud Text-to-Speech. `https://cloud.google.com/text-to-speech/docs/gemini-tts`. Accessed: Aug. 9, 2026.

Alex Graves, Santiago Fernández, Faustino Gomez, and Jürgen Schmidhuber. 2006. Connectionist temporal classification: labelling unsegmented sequence data with recurrent neural networks. In *Proceedings of the 23rd international conference on Machine learning*, pages 369–376.

Anmol Gulati, James Qin, Chung-Cheng Chiu, Niki Parmar, Yu Zhang, Jiahui Yu, Wei Han, Shibo Wang, Zhengdong Zhang, Yonghui Wu, and Ruoming Pang. 2020. Conformer: Convolution-augmented Transformer for Speech Recognition. In *Interspeech 2020*, pages 5036–5040.

Injy Hamed, Nizar Habash, Slim Abdennadher, and Ngoc Thang Vu. 2022. ArzEn-ST: A three-way speech translation corpus for code-switched Egyptian Arabic-English. In *Proceedings of the Seventh Arabic Natural Language Processing Workshop (WANLP)*, pages 119–130, Abu Dhabi, United Arab Emirates (Hybrid). Association for Computational Linguistics.

Injy Hamed, Ngoc Thang Vu, and Slim Abdennadher. 2020. ArzEn: A speech corpus for code-switched Egyptian Arabic-English. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 4237–4246, Marseille, France. European Language Resources Association.

Kenneth Heafield. 2011. KenLM: Faster and smaller language model queries. In *Proceedings of the Sixth Workshop on Statistical Machine Translation*, pages 187–197, Edinburgh, Scotland. Association for Computational Linguistics.

Geoffrey Hinton, Li Deng, Dong Yu, George E. Dahl, Abdel-rahman Mohamed, Navdeep Jaitly, Andrew Senior, Vincent Vanhoucke, Patrick Nguyen, Tara N. Sainath, and Brian Kingsbury. 2012. Deep neural networks for acoustic modeling in speech recognition: The shared views of four research groups. *IEEE Signal Processing Magazine*, 29(6):82–97.

Wei-Ning Hsu, Benjamin Bolte, Yao-Hung Hubert Tsai, Kushal Lakhotia, Ruslan Salakhutdinov, and Abdelrahman Mohamed. 2021. HuBERT: Self-supervised speech representation learning by masked prediction of hidden units. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 29:3451–3460.

Musab Iskandar, Mohamed Ebrahim, Adel Alharbi, Khaled Alharbi, and Youssef Mohamed. 2026. AI Elites at NADI 2026 Shared Task: Selecting on Out-of-Domain Accuracy for Cross-Domain Spoken Arabic Dialect Identification. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

KAND CA. 2026. Lahgtna huggingface dataset.

Woohyun Kang, Md Jahangir Alam, and Abderrahim Fathan. 2022. Deep learning-based end-to-end spoken language identification system for domain-mismatched scenario. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 7339–7343, Marseille, France. European Language Resources Association.

Salma Khaled. 2026. CIS-OpenSLU at NADI 2026 Shared Task: ASR-Matched Training and Confidence-Based Rejection for Open-Set Tunisian Arabic Intent Recognition. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Ahmed Khamis and Hesham Ali Ahmed. 2026. LLM-to-Speech: A synthetic data pipeline for training dialectal text-to-speech models. In *Proceedings of the 2nd Workshop on NLP for Languages Using Arabic Script*, page 47–54. Association for Computational Linguistics.

Ondřej Klejch, William Lamb, and Peter Bell. 2025. A Practitioner's Guide to Building ASR Models for Low-Resource Languages: A Case Study on Scottish Gaelic. In *Interspeech 2025*, pages 728–732.

Moksh Kopikar and Naman Mandloi. 2026. Amchi-low resource language ASR with crowdsourced post-processing. In *Proc. SpeechProsody 2026*, pages 924–925.

Saurabh Kumar, Amartyaveer, and Prasanta Kumar Ghosh. 2025. Jointly Improving Dialect Identification and ASR in Indian Languages using Multimodal Feature Fusion. In *Interspeech 2025*, pages 2770–2774.

Imen Laouirine, Rami Kammoun, and Fethi Bougares. 2024. TunArTTS: Tunisian Arabic text-to-speech corpus. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 16879–16889, Torino, Italia. ELRA and ICCL.

Gaëlle Laperrière, Valentin Pelloin, Antoine Caubrière, Salima Mdhaffar, Nathalie Camelin, Sahar Ghannay, Bassam Jabaian, and Yannick Estève. 2022. The spoken language understanding MEDIA benchmark dataset in the era of deep learning: data updates, training and evaluation tools. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 1595–1602, Marseille, France. European Language Resources Association.

Antoine Laurent, Souhir Gahbiche, Ha Nguyen, Haroun Elleuch, Fethi Bougares, Antoine Thiol, Hugo Riguidel, Salima Mdhaffar, Gaëlle Laperrière, Lucas Maison, Sameer Khurana, and Yannick Estève. 2023. ON-TRAC consortium systems for the IWSLT 2023 dialectal and low-resource speech translation tasks. In *Proceedings of the 20th International Conference on Spoken Language Translation (IWSLT 2023)*, pages 219–226, Toronto, Canada (in-person and online). Association for Computational Linguistics.

Beomseok Lee, Ioan Calapodescu, Marco Gaido, Matteo Negri, and Laurent Besacier. 2024. Speech-MASSIVE: A Multilingual Speech Dataset for SLU and Beyond. In *Interspeech 2024*, pages 817–821.

Yooyoung Lee, Craig Greenberg, Lisa Mason, and Elliot Singer. 2022. NIST 2022 language recognition evaluation plan. National Institute of Standards and Technology.

Chahira Lhioui, Anis Zouaghi, and Mounir Zrigui. 2013. A combined method based on stochastic and linguistic paradigm for the understanding of Arabic spontaneous utterances. In *International Conference on Intelligent Text Processing and Computational Linguistics*, pages 549–558. Springer.

Zhaolin Li and Jan Niehues. 2025. In-context Language Learning for Endangered Languages in Speech Recognition. In *Interspeech 2025*, pages 738–742.

Mohamed Lichouri, Mourad Abbas, Rachida Djeradi, and Amar Djeradi. 2019. An Arabic multi-domain spoken language understanding system. In *Proceedings of the 3rd International Conference on Natural Language and Speech Processing*, pages 49–53.

Shervin Malmasi and Marcos Zampieri. 2016. Arabic dialect identification in speech transcripts. In *Proceedings of the Third Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial3)*, pages 106–113, Osaka, Japan. The COLING 2016 Organizing Committee.

Salima Mdhaffar, Fethi Bougares, Renato de Mori, Salah Zaiem, Mirco Ravanelli, and Yannick Estève. 2024a. TARIC-SLU: A Tunisian benchmark dataset for spoken language understanding. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 15606–15616, Torino, Italia. ELRA and ICCL.

Salima Mdhaffar, Haroun Elleuch, Fethi Bougares, and Yannick Estève. 2024b. Performance analysis of speech encoders for low-resource SLU and ASR in Tunisian dialect. In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 130–139, Bangkok, Thailand. Association for Computational Linguistics.

Salima Mdhaffar, Haroun Elleuch, Chaimae Chellaf, Ha Nguyen, and Yannick Estève. 2025. Sense models: an open source solution for multilingual and multimodal semantic-based tasks. In *2025 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, pages 1–8. IEEE.

Pouya Mehralian and Hugo Van hamme. 2025. Leveraging geographic metadata for dialect-aware speech recognition. In *Interspeech 2025*, pages 1153–1157.

Gabriel Mittag and Sebastian Möller. 2020. Deep Learning Based Assessment of Synthetic Speech Naturalness. In *Interspeech 2020*, pages 1748–1752.

Gabriel Mittag, Babak Naderi, Assmaa Chehadi, and Sebastian Möller. 2021. Nisqa: A deep cnn-self-attention model for multidimensional speech quality prediction with crowdsourced datasets. In *Interspeech 2021*, page 2127–2131. ISCA.

Hamid Mojarad and Kevin Tang. 2025. Automatic Speech Recognition of African American English: Lexical and Contextual Effects. In *Interspeech 2025*, pages 3883–3887.

Hamdy Mubarak, Amir Hussein, Shammur Absar Chowdhury, and Ahmed Ali. 2021. QASR: QCRI Aljazeera speech resource a large scale annotated Arabic speech corpus. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 2274–2285, Online. Association for Computational Linguistics.

oddadmix. 2024. lahgtna-chatterbox-v1. Hugging Face. Accessed: Aug. 9, 2026.

Omnilingual ASR, Gil Keren, Artyom Kozhevnikov, Yen Meng, Christophe Ropers, Matthew Setzler, Skyler Wang, Ife Adebara, Michael Auli, Can Balioglu, and 1 others. 2025. Omnilingual ASR: Open-source multilingual speech recognition for 1600+ languages. *arXiv preprint arXiv:2511.09690*.

Abdulaziz Otaif, Hazem Bakhshwain, Abdulhamid Aldoobi, Meshal Alamr, and Abdullah Aldahlawi. 2026. Thakaa at NADI 2026 Shared Task: ROVER Ensembling and Multi-Representation Fusion for Robust

Arabic ASR and Dialect Identification. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pages 311–318, Philadelphia, Pennsylvania, USA. Association for Computational Linguistics.

Daniel S. Park, William Chan, Yu Zhang, Chung-Cheng Chiu, Barret Zoph, Ekin D. Cubuk, and Quoc V. Le. 2019. SpecAugment: A Simple Data Augmentation Method for Automatic Speech Recognition. In *Interspeech 2019*, pages 2613–2617.

Karol J Piczak. 2015. ESC: Dataset for environmental sound classification. In *Proceedings of the 23rd ACM international conference on Multimedia*, pages 1015–1018.

Maja Popović. 2015. chrF: character n-gram F-score for automatic MT evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation*, pages 392–395, Lisbon, Portugal. Association for Computational Linguistics.

Matt Post. 2018. A call for clarity in reporting BLEU scores. In *Proceedings of the Third Conference on Machine Translation: Research Papers*, pages 186–191, Brussels, Belgium. Association for Computational Linguistics.

Jeena Prakash, Blessingh Kumar, Kadri Hacioglu, Bidisha Sharma, Sindhuja Gopalan, Malolan Chetlur, Shankar Venkatesan, and Andreas Stolcke. 2025. Better Pseudo-labeling with Multi-ASR Fusion and Error Correction by SpeechLLM. In *Interspeech 2025*, pages 579–583.

Vineel Pratap, Andros Tjandra, Bowen Shi, Paden Tomasello, Arun Babu, Sayani Kundu, Ali Elkahky, Zhaoheng Ni, Apoorv Vyas, Maryam Fazel-Zarandi, and 1 others. 2024. Scaling speech technology to 1,000+ languages. *Journal of Machine Learning Research*, 25(97):1–52.

Lawrence R. Rabiner. 1989. A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. Robust speech recognition via large-scale weak supervision. In *International conference on machine learning*, pages 28492–28518. PMLR.

Dima Rekesh, Nithin Rao Koluguri, Samuel Kriman, Somshubra Majumdar, Vahid Noroozi, He Huang, Oleksii Hrinchuk, Krishna Puvvada, Ankur Kumar, Jagadeesh Balam, and 1 others. 2023. Fast Conformer with linearly scalable attention for efficient speech recognition. In *2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, pages 1–8. IEEE.

Mohamed Sabaa and Amr Sabaa. 2026. Sabaa at NADI 2026 Subtask 1.1: Matching Annotation Conventions for Dialectal Arabic ASR. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Omar Samir, Youssef Waleed, Youssef Tamer, and Amir Mohamed. 2024. Fine-tuning XTTS V2 for Egyptian Arabic.

Dana Serditova, Kevin Tang, and Jochen Steffens. 2025. Automatic speech recognition biases in Newcastle English: An error analysis. *arXiv preprint arXiv:2506.16558*.

Tajwaar Shafiq, Hunzalah Hassan Bhatti, Firoj Alam, and Shammur Absar Chowdhury. 2026. Aslema at NADI 2026: Data Augmentation for Intent Recognition and Slot Filling. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Omar Shoaib, Mohamed Sokar, Mohamed Motawie, Omar AboElFottouh, Ahmed Hassouna, AlAmir Hassan, Nader Essam, and Wael Ali. 2026. Resonate at NADI 2026 Shared Task: Robust Speech Recognition and Dialect Identification under Acoustic and Domain Shift. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Suwon Shon, Ahmed Ali, and James Glass. 2017. MIT-QCRI Arabic dialect identification system for the 2017 multi-genre broadcast challenge. In *2017 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, pages 374–380. IEEE.

Silero Team. 2024. Silero VAD: pre-trained enterprise-grade voice activity detector (VAD), number detector and language classifier. `https://github.com/snakers4/silero-vad`.

SILMA AI. 2026. Arabic Text-to-Speech (TTS) models. `https://silma.ai/arabic-tts-models`. Accessed: Aug. 9, 2026.

Marta Smolarek. 2019. Amazon Polly adds Arabic language support. AWS Machine Learning Blog. Accessed: Aug. 9, 2026.

David Snyder, Daniel Garcia-Romero, Alan McCree, Gregory Sell, Daniel Povey, and Sanjeev Khudanpur. 2018. Spoken language recognition using x-vectors. In *Odyssey*, volume 2018, pages 105–111.

Peter Sullivan, AbdelRahim Elmadany, and Muhammad Abdul-Mageed. 2023. On the robustness of Arabic speech dialect identification. In *Interspeech 2023*, pages 5326–5330.

Peter Sullivan, AbdelRahim A. Elmadany, Alcides Alcoba Inciarte, and Muhammad Abdul-Mageed. 2026. Arab voices: Mapping standard and dialectal Arabic speech technology. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 11843–11878, San Diego, California, United States. Association for Computational Linguistics.

Bashar Talafha, Karima Kadaoui, Samar Mohamed Magdy, Mariem Habiboullah, Chafei Mohamed Chafei, Ahmed Oumar El-Shangiti, Hiba Zayed, Mohamedou Cheikh Tourad, Rahaf Alhamouri, Rwaa Assi, Aisha Alraeesi, Hour Mohamed, Fakhraddin Alwajih, Abdelrahman Mohamed, Abdellah El Mekki, El Moatez Billah Nagoudi, Benelhadj Djelloul Mama Saadia, Hamzah A. Alsayadi, Walid Al-Dhabyani, and 8 others. 2024. Casablanca: Data and models for multidialectal Arabic speech recognition. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 21745–21758, Miami, Florida, USA. Association for Computational Linguistics.

Bashar Talafha, Hawau Olamide Toyin, Peter Sullivan, AbdelRahim A. Elmadany, Abdurrahman Juma, Amirbek Djanibekov, Chiyu Zhang, Hamad Alshehhi, Hanan Aldarmaki, Mustafa Jarrar, Nizar Habash, and Muhammad Abdul-Mageed. 2025. NADI 2025: The first multidialectal Arabic speech processing shared task. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 720–733, Suzhou, China. Association for Computational Linguistics.

Xu Tan, Tao Qin, Frank Soong, and Tie-Yan Liu. 2021. A survey on neural speech synthesis. *Preprint*, arXiv:2106.15561.

Hawau Olamide Toyin, Amirbek Djanibekov, Ajinkya Kulkarni, and Hanan Aldarmaki. 2023. ArTST: Arabic text and speech transformer. In *Proceedings of ArabicNLP 2023*, pages 41–51, Singapore (Hybrid). Association for Computational Linguistics.

Gokhan Tur and Renato De Mori. 2011. *Spoken language understanding: Systems for extracting semantic information from speech*. John Wiley & Sons.

Jörgen Valk and Tanel Alumäe. 2021. Voxlingua107: a dataset for spoken language recognition. In *2021 IEEE Spoken Language Technology Workshop (SLT)*, pages 652–658. IEEE.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems*, 30.

Pengcheng Wang, Sheng Li, and Takahiro Shinozaki. 2025a. RAG-Boost: Retrieval-Augmented Generation Enhanced LLM-based Speech Recognition. In *Workshop on Multilingual Conversational Speech Language Model (MLC-SLM)*, pages 54–55.

Yingzhi Wang, Anas Alhmoud, and Muhammad Alqurishi. 2025b. Open Universal Arabic ASR Leaderboard. In *Interspeech 2025*, pages 1168–1172.

Ahmed Wasfy and Nourhan Anber. 2026. KAND CA at NADI 2026 Shared Task: Multi-Stage Fine-Tuning for Spoken Arabic Dialects. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Bajian Xiang, Cheng Wen, Han Zhao, Hao Wang, Haoxu Wang, Jiawei Jin, Jiayan Cui, Jie Chen, Mengxi Nie, Tianyu Zhao, Weiqin Li, Xiang Lv, Xiangang Li, Yang Xiang, and Yang Zhou. 2026. Qwen-Audio-3.0-TTS: Freely controllable and highly robust speech synthesis with multi-stage training paradigm. *Preprint*, arXiv:2607.23938.

Tianyi Xu, Hongjie Chen, Qing Wang, Lv Hang, Jian Kang, Jie Li, Zhennan Lin, Yongxiang Li, and Lei Xie. 2025. Leveraging LLM and Self-Supervised Training Models for Speech Recognition in Chinese Dialects: A Comparative Analysis. In *Interspeech 2025*, pages 584–588.

Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. 2021. mT5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 483–498, Online. Association for Computational Linguistics.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, and 1 others. 2025a. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Hongli Yang, Yizhou Peng, Hao Huang, and Sheng Li. 2025b. Adapting Whisper for Parameter-efficient Code-Switching Speech Recognition via Soft Prompt Tuning. In *Interspeech 2025*, pages 5203–5207.

Sarah Yassine and Abdulkadir Shehu Bichi. 2026. CodeZone Research Group at NADI 2026 Shared Task: Cross-Domain Score Calibration for Spoken Arabic Dialect Identification. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Shengkui Zhao, Zexu Pan, and Bin Ma. 2025. ClearerVoice-Studio: Bridging Advanced Speech Processing Research and Practical Deployment. In *Interspeech 2025*, pages 2980–2984.

Yixuan Zhou, Guoyang Zeng, Xin Liu, Xiang Li, Renjie Yu, Jiancheng Gui, Jiaheng Wu, Ziyang Wang, Xudong Shen, Runchuan Ye, and 1 others. 2026. VoxCPM2 technical report. *arXiv preprint arXiv:2606.06928*.

Han Zhu, Lingxuan Ye, Wei Kang, Zengwei Yao, Liyong Guo, Fangjun Kuang, Zhifeng Han, Weiji Zhuang, Long Lin, and Daniel Povey. 2026. OmniVoice: Towards omnilingual zero-shot text-to-speech with diffusion language models. *arXiv preprint arXiv:2604.00688*.

Jianheng Zhuo, Yifan Yang, Yiwen Shao, Yong Xu, Dong Yu, Kai Yu, and Xie Chen. 2025. VietASR: Achieving Industry-level Vietnamese ASR with 50-hour labeled data and Large-Scale Speech Pretraining. In *Interspeech 2025*, pages 1163–1167.

Ömer Tarik Özyilmaz, Matt Coler, and Matias Valdenegro-Toro. 2025. Overcoming Data Scarcity in Multi-Dialectal Arabic ASR via Whisper Fine-Tuning. In *Interspeech 2025*, pages 1158–1162.

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A Comprehensive Literature Review

A.1 Automatic Speech Recognition

Foundations of ASR. Automatic speech recognition has progressed from statistical HMM-GMM systems (Rabiner, 1989) and hybrid DNN-HMM acoustic models (Hinton et al., 2012) to end-to-end architectures trained with connectionist temporal classification (Graves et al., 2006) and attention-based encoder-decoders (Chan et al., 2016). The Conformer (Gulati et al., 2020) combined convolution and self-attention as a standard encoder backbone, while self-supervised pretraining (Baevski et al., 2020; Hsu et al., 2021) and large-scale weakly supervised training (Radford et al., 2023) substantially reduced the field's dependence on labeled data. These advances underpin nearly all recent dialectal and low-resource ASR work, including on Arabic, where Whisper- and MMS-based models (Radford et al., 2023; Pratap et al., 2024) now serve as the default starting point for dialect-specific adaptation (Wang et al., 2025b; Ömer Tarik Özyilmaz et al., 2025).

Low-Resource and Multilingual ASR. A substantial line of work targets languages and varieties with limited labeled data. Approaches include transfer learning from phonetically similar high-resource languages combined with community-sourced lexical resources (Kopikar and Mandloi, 2026), self-supervised pretraining on large unlabeled corpora followed by minimal fine-tuning (Zhuo et al., 2025), and hybrid HMM/self-supervised pipelines that outperform naïvely fine-tuned end-to-end models (Klejch et al., 2025). In-context learning with LLMs has also been shown to acquire unseen low-resource languages without dedicated ASR training (Li and Niehues, 2025), and self-supervised encoders have been scaled to over a thousand African languages (Alabi et al., 2024). Arabic dialects sit squarely within this low-resource paradigm: despite being spoken by hundreds of millions of people, most dialectal varieties lack the transcribed corpora available for MSA (Ali et al., 2017; Bouamor et al., 2018), motivating dedicated efforts such as the NADI shared task series (Talafha et al., 2025).

Dialectal, Accented, and Sociolinguistic Robustness. ASR systems trained predominantly on standard or high-resource varieties degrade substantially on dialectal and accented speech. This has been documented for German dialects (Blaschke et al., 2025), Chinese dialects (Xu et al., 2025), regional English varieties such as Newcastle English (Serditova et al., 2025), and African American English (Mojarad and Tang, 2025). Proposed mitigations range from geographic-metadata conditioning (Mehralian and Van hamme, 2025) to mixtures of accent-specific LoRA experts (Bagat et al., 2025). Arabic exhibits some of the most pronounced dialectal variation of any widely spoken language, and a growing body of work specifically fine-tunes and benchmarks multilingual ASR models across Arabic dialects (Wang et al., 2025b; Ömer Tarik Özyilmaz et al., 2025), jointly modeling dialect identification alongside recognition (Kumar et al., 2025). Code-switching compounds this variation further, and while most code-switching ASR work has focused on non-Arabic language pairs (Biswas et al., 2025; Yang et al., 2025b), Arabic speech routinely mixes dialect, MSA, and other languages, a setting NADI 2026 targets directly.

Speech-LLM Integration and Error Correction. Recent work couples ASR with large language models for contextual biasing (Gong

et al., 2025), retrieval-augmented correction (Wang et al., 2025a), and multi-ASR ensemble arbitration (Prakash et al., 2025), extending classic hypothesis-fusion approaches with LLM-based reasoning. These techniques are largely language-agnostic and have primarily been validated on high-resource languages; their application to dialectal Arabic, where named entities, code-switched terms, and non-standard orthography are common, remains comparatively unexplored.

A.2 Spoken Language and Dialect ID

Spoken language identification and its cousin spoken dialect identification have largely transitioned from traditional approaches such as i-vector (Dehak et al., 2011) and x-vector (Snyder et al., 2018) to end-to-end deep neural net models often using a transformer (Vaswani et al., 2017; Babu et al., 2022) or ECAPA-TDNN (Desplanques et al., 2020) backbone. Arabic dialect identification has similarly followed these trends moving from i-vector approaches during the MGB-3 shared task (Ali et al., 2017; Shon et al., 2017), to end-to-end neural nets trained from scratch in the follow-up MGB-5 shared task (Ali et al., 2019), and finetuning pretrained models for dialect identification in NADI 2025 (Elleuch et al., 2025; Talafha et al., 2025).

One consistent issue with spoken language identification is the problem of cross domain performance (Sullivan et al., 2023; Abdullah et al., 2025; Kang et al., 2022; Dey et al., 2023). Models may overfit to training domains, rendering them ineffective in real-world scenarios.

A.3 Text-to-Speech.

Driven by recent breakthroughs in autoregressive, diffusion-based, and zero-shot architectures, modern TTS systems have demonstrated remarkable naturalness and expressiveness (Tan et al., 2021). However, while high-resource languages—and models such as Gemini-TTS (Google Cloud, 2026) or Qwen-based speech architectures (Xiang et al., 2026)—have advanced significantly, progress in the Arabic language remains constrained. The primary bottleneck lies in orthographic ambiguity (e.g., the omission of diacritics) and the rich morphological and dialectal variations across the Arab world. To provide a structured overview of existing Arabic TTS capabilities, we categorize existing speech synthesis systems into three primary groups: multilingual models, MSA models, and DA models.

Multilingual Models Broad-coverage foundational speech models, such as XTTS (Casanova et al., 2024), SeamlessM4T (Barrault et al., 2023a), SpeechT5 (Ao et al., 2022), and ElevenLabs Multilingual v2 (ElevenLabs, 2023), support Arabic within unified cross-lingual representations. However, they struggle with Arabic phonetic subtleties, orthographic ambiguity, and dialect-specific prosody. Furthermore, because these models do not allow explicit conditioning on specific regional dialects, their ability to synthesize authentic local varieties remains limited-highlighting a critical gap in existing multilingual TTS architectures.

MSA Models Dedicated acoustic and neural synthesis architectures trained specifically on diacritized formal corpora. Representative examples include ArTST (Toyin et al., 2023), and SILMA TTS MSA (SILMA AI, 2026), alongside commercial engines like Amazon Polly (*Zeina*) (Smolarek, 2019). While these models achieve high intelligibility and naturalness in formal registers and news broadcast domains, they fail to generalize to informal, colloquial speech patterns due to distinct dialectal vocabulary, syntax, and phonology.

DA Models Emerging architectures and fine-tuned models specifically tailored to capture country- and region-specific phonetic, rhythmic, and prosodic characteristics. Notable examples include NileTTS (Khamis and Ahmed, 2026), EGTTS (Samir et al., 2024), Habibi-TTS (Chen et al., 2026), and lahgtna-chatterbox-v1 (oddadmix, 2024), which target regional varieties such as Egyptian, Levantine, and Gulf Arabic. Despite their necessity for realistic conversational AI, dialectal TTS systems remain severely under-resourced compared to English and MSA due to the scarcity of high-quality, paired dialectal speech-text corpora.

A.4 Spoken Language Translation

Multilingual Models Large-scale multilingual speech translation models provide the most developed approach for Arabic SLT. Representative systems include SeamlessM4T (Barrault et al., 2023a) and Whisper (Radford et al., 2023), which benefit from multilingual speech pretraining and can support Arabic as either a source or target language. More recent foundation models such as SeamlessM4T v2 (Barrault et al., 2023b) further improve multilingual speech-to-text and speech-to-speech translation capabilities. However, these models are primarily optimized for broad multilin-

gual coverage and may struggle to preserve dialect-specific lexical, phonological, and prosodic characteristics, especially for low-resource Arabic varieties.

DA Models Spoken translation for dialectal Arabic remains considerably under-resourced compared with MSA and high-resource languages. Recent efforts have increasingly focused on developing speech–translation resources for Arabic dialects. In particular, the WhiteHouse corpus (Bougares et al., 2026) was recently introduced as the first multi-dialectal Arabic–English speech translation corpus, providing three-way parallel speech, transcription, and English translation data across multiple Arabic dialects. For Tunisian Arabic, the IWSLT Dialectal Speech Translation shared task has provided dedicated resources for Tunisian–English speech translation, including 160 hours (Abdulmumin et al., 2025). These resources have enabled the development and evaluation of dialect-specific speech translation systems (Chellaf et al., 2025; Laurent et al., 2023), while challenges related to dialectal variation, code-switching, spontaneous speech, and limited parallel supervision remain largely open. In addition, TEDxTN (Bougares et al., 2025) is a three-way code-switched Tunisian Arabic–English speech translation corpus comprising 108 TEDx talks and approximately 25 hours of speech from speakers representing more than 11 regions of Tunisia. The corpus provides manually validated transcriptions and English translations at the utterance level, making it a valuable resource for studying code-switched and regionally diverse Tunisian speech translation.

For Egyptian Arabic, ArzEnST (Hamed et al., 2022) is a three-way Egyptian Arabic–English code-switched speech translation corpus derived from the ArzEn speech corpus (Hamed et al., 2020). It consists of approximately 12 hours of speech and 6,216 sentences collected through informal interviews with bilingual speakers. The corpus provides speech, Egyptian Arabic transcriptions, and English translations, enabling the development of cascaded speech translation systems in which an ASR model first transcribes the speech and an MT model subsequently translates the resulting transcripts.

A.5 Spoken Language Understanding

Driven by recent advances in pretrained speech encoders, multilingual foundation models, and end-to-end architectures, modern Spoken Language Understanding (SLU) systems have increasingly moved beyond cascaded ASR and Natural Language Understanding (NLU) pipelines toward end-to-end models. However, while high-resource languages benefit from large-scale annotated benchmarks and powerful pretrained models, Arabic SLU remains comparatively under-resourced, particularly for dialectal varieties.

MSA Models Existing MSA SLU systems range from conventional statistical and neural approaches to recent pretrained Transformer-based models. Early Arabic SLU work relied on classical classifiers such as Multinomial Naïve Bayes, Logistic Regression, Support Vector Machines, and CNNs for intent identification (Lichouri et al., 2019). For slot filling, earlier Arabic SLU systems have employed rule-based, statistical, and sequence-labeling approaches to identify semantic concepts and their relationships within user utterances (Lhioui et al., 2013). A major resource for evaluating speech-based MSA SLU is Speech-MASSIVE (Lee et al., 2024), which extends the MASSIVE benchmark (FitzGerald et al., 2023) to the speech modality and includes Arabic among its supported languages. The benchmark covers 18 domains, 60 intents, and 55 slot types, enabling the evaluation of both intent classification and slot filling from speech. However, the Arabic subset remains extremely limited, with only 115 training utterances, making it more suitable for few-shot and cross-lingual evaluation than for training dedicated MSA SLU models. This scarcity of annotated speech-semantic data remains one of the main challenges for advancing MSA SLU.

DA Models Recent work on dialectal Arabic SLU has mainly focused on low-resource Tunisian Arabic. (Mdhaffar et al., 2024b) benchmarked a range of pretrained speech encoders for end-to-end SLU on TARIC-SLU (Mdhaffar et al., 2024a), including monolingual models such as wav2vec 2.0 (Baevski et al., 2020), HuBERT (Hsu et al., 2021), WavLM (Chen et al., 2022), and data2vec 2.0 (Baevski et al., 2023), as well as multilingual models such as XLS-R (Babu et al., 2022), MMS (Pratap et al., 2024), MMS-1B (Pratap et al., 2024), and w2v-BERT 2.0 (Barrault et al., 2023b). Among the evaluated systems, w2v-BERT 2.0 achieved the best overall SLU performance. More recently, semantically enhanced models such as SENSE (Mdhaffar et al., 2025), which incorpo-

rate textual semantic information through a teacher–student training strategy, have been investigated for Tunisian SLU and have demonstrated further performance improvements. More recently, SLURP-TN (Elleuch et al., 2026a) was introduced as a new resource for Tunisian Arabic SLU, providing speech-level intent and slot annotations for task-oriented conversational scenarios.

B Additional Task Information

B.1 TTS Data Overview

Following extraction, the text underwent a structured preprocessing pipeline. We normalized Unicode representations (NFKC), removed diacritics, Tatweel, emojis, parenthetical metadata, HTML tags, and non-Arabic characters. We further normalized character runs (reducing repeated characters and removing online laughter), stripped non-standard punctuation, and enforced strict quality constraints by removing sentences containing English words, URLs, user tags, long gibberish words, or fewer than 5 words. In total, the benchmark comprises 4,001 samples spanning the 10 target dialects, sourced from public datasets, web data, and private team resources. The country-wise distribution of the evaluation samples is detailed in Table B.1.

B.2 SLT Data Overview

We provide an overview of the duration and word counts for the SLT subtask in Table B.2.

B.3 SLU Data Overview

The evaluation material was recorded by 12 speakers between 23 and 28 years of age. The speaker distribution is exactly balanced, with six male and six female speakers. At the utterance level, male speakers contributed 523 utterances and female speakers contributed 466, corresponding to 52.88% and 47.12% of the evaluation set, respectively. The speakers represent five regional backgrounds. Five speakers are associated with Greater Tunis, four with Medenine, and one each with Kebili, Kairouan, and Sfax. These speakers contributed 492, 322, 110, 41, and 24 utterances, respectively. Although the regional distribution is not uniform, it introduces variation beyond a single urban variety and preserves the broader regional motivation of SLURP-TN. A central design choice was to combine domains already represented in SLURP-TN with previously unseen domains. Emails, Weather, and General constitute the three returning domains. Calendar, Question Answering, Music, and Cooking constitute four new domains. These domains are absent from the SLURP-TN training and development sets provided to participants. They therefore support zero-shot evaluation at the domain level: systems must process utterances belonging to application domains that were not represented during system development. The official evaluation set consequently assesses two complementary forms of generalisation. For the returning domains, systems must generalise to new speakers and previously unseen utterances while operating within familiar application domains. For the zero-shot domains, systems must additionally transfer their learned acoustic, lexical, and semantic representations to new application areas.

C Baseline Systems

Task 1 ASR: For all three of the ASR subtasks we use `whisper-Large-v3` (Radford et al., 2023), a multilingual ASR model trained on a noisy-labeled internet audio. During the models training, there was no distinguishing between varieties of Arabic, however, it can be assumed that by training off of subtitles, there may be a preference for generating MSA.

Task 2 Spoken Dialect Identification: We adopt the baseline from NADI 2025 ADI subtask (Talafha et al., 2025), with a number of modifications. It starts with a ECAPA-TDNN system [8] originally trained on LinguaVox107 (Valk and Alumäe, 2021), which we then finetune on a 10hr-per-dialect version of ADI-20 dataset.

We use a learning rate schedule as follows starting with only the final classification layer thawed: 9k steps linear warm up from $3.3 * 10^{-5}$ to 10^{-4}, 6k steps at constant LR, thaw remaining layers, 15k steps linear warm up from $3.3 * 10^{-5}$ to 10^{-4}, 30k steps constant LR, and a final 30k steps with exponential decay.

Task 3 Text-To-Speech: We used the XTTS-v2 model as the baseline for this subtask to generate audio samples across all dialects. A single reference speaker, "Suad Qasim," was selected and used consistently across all dialects. XTTS is a Cross-Lingual TTS model developed by

[8] https://huggingface.co/speechbrain/lang-id-voxlingua107-ecapa

Comparison Dimension	2020	2021	2022	2023	2024	2025	2026
Edition Overview							
Modality	Text	Text	Text	Text	Text	Speech	Speech
# Subtasks	2	4	2	3	3	3	8
Task families	T1	T1	T1, T2	T1, T3	T1, T3, T4	T1, T5, T6	T1, T5, T7–T9
Task Coverage							
Dialect Identification (T1)	✓	✓	✓	✓	✓	✓	✓
Sentiment Analysis (T2)	–	–	✓	–	–	–	–
Dialect-to-MSA MT (T3)	–	–	–	✓	✓	–	–
ALDi Estimation (T4)	–	–	–	–	✓	–	–
Automatic Speech Recognition (T5)	–	–	–	–	–	✓	✓
Diacritic Restoration (T6)	–	–	–	–	–	✓	–
Text-to-Speech (T7)	–	–	–	–	–	–	✓
Spoken Language Translation (T8)	–	–	–	–	–	–	✓
Spoken Language Understanding (T9)	–	–	–	–	–	–	✓
Task Characteristics							
Province-level DID	✓	✓	–	–	–	–	–
MSA/DA Geographic DID	–	✓	–	–	–	–	–
Multi-label DID	–	–	–	–	✓	–	–
Robust / Low-bandwidth ASR	–	–	–	–	–	–	✓
Mixed-dialect ASR	–	–	–	–	–	–	✓
Code-switched ASR	–	–	–	–	–	–	✓
Out-of-domain SDID	–	–	–	–	–	–	✓
Zero-shot Domain Evaluation	–	–	–	–	–	–	✓
Open/closed Tracks	–	–	–	C/O	C/O	C/O (T6)	O (T7)
Geographic / Variety Scope							
DI Coverage	21 countries	21 countries	18 countries	18 countries	Country-level	8 dialects	6 countries 18 city dialects
Province-level Coverage	100 provinces	100 provinces	–	–	–	–	–
MT / SLT Coverage	–	–	–	4 dialects	4 dialects	–	8 dialects
TTS Coverage	–	–	–	–	–	–	10 countries
SLU Coverage	–	–	–	–	–	–	Tunisian
Robust ASR Coverage	–	–	–	–	–	–	Low-bandwidth / degraded speech
Mixed-dialect ASR Coverage	–	–	–	–	–	–	6 countries >15 subdialects
Code-switched ASR Coverage	–	–	–	–	–	–	Tunisian + EN/FR
Evaluation							
Metric families	M1	M1	M1, M2	M1, M3	M1, M3, M4	M5–M8	M3, M5–M14
Participation and Impact							
Registered teams	61	53	41	58	51	44	65
Participating teams	18	8	21	18	12	8	21
Test submissions	56	68	105	76	76	100	48
Accepted papers	14	7	15	13	8	7	14

Table A.1: Detailed evolution of the NADI shared task series from 2020 to 2026. The series focused on text-based tasks from 2020 to 2024, transitioned to speech in 2025, and further expanded its speech-processing scope in 2026. Task-family codes are normalized across editions and do not correspond to the original task numbering within individual NADI editions. T1: Dialect Identification; T2: Sentiment Analysis; T3: Dialect-to-MSA Machine Translation; T4: Arabic Level of Dialectness Estimation; T5: Automatic Speech Recognition; T6: Diacritic Restoration; T7: Text-to-Speech; T8: Spoken Language Translation; and T9: Spoken Language Understanding. Metric-family codes are normalized across editions. M1: Macro-F1; M2: F1-PN; M3: BLEU; M4: Root Mean Squared Error; M5: Accuracy; M6: Average Cost (Cavg); M7: Word Error Rate; M8: Character Error Rate; M9: UTMOSv2; M10: NISQA; M11: chrF; M12: Weighted F1; M13: Concept Error Rate; M14: Concept/Value Error Rate. DI: Dialect Identification; MSA: Modern Standard Arabic; DA: Dialectal Arabic; SDI: Spoken Dialect Identification; ASR: Automatic Speech Recognition; MT: Machine Translation; TTS: Text-to-Speech; SLT: Spoken Language Translation; SLU: Spoken Language Understanding; ALDi: Arabic Level of Dialectness; C/O: Closed/Open; EN: English; FR: French. For task-property rows, a check mark indicates that the property is present, whereas a dash (–) indicates that it is absent. For numeric rows, a dash indicates unavailable or unreported information.

Coqui [9] that can synthesize speech from text while using a short reference audio clip to reproduce characteristics of a target speaker's voice. While this baseline was not available for comparison during the original evaluation period, we provide it

[9] https://docs.coqui.ai/en/latest/index.html

Subtask	Codabench Participants	Dev Submissions	Test Submissions
1.1 Robust ASR	30	5	9
1.2 Multidialect ASR	19	2	4
1.3 CS ASR	17	4	6
2 SDID	39	6	14*
3 TTS	N/A	N/A	1
4 SLT	15	3	2
5.1 Intent Classification	16	2	8†
5.2 Slot Filling	8	0	4‡

Table A.2: Overview of participation across the eight subtasks. We count the number of people signed up for each Codabench and number of teams that publicly submitted for both the Dev and Test phase. Test submissions counts include: ∗ two duplicate submissions; † one unregistered and two duplicate submissions; ‡ two invalid submissions. For the final leaderboard only the top performing submission for each team is included.

Country Code	Country Name	Sample Count
ae	United Arab Emirates	537
dz	Algeria	293
eg	Egypt	424
jo	Jordan	631
ma	Morocco	201
ps	Palestine	750
sa	Saudi Arabia	100
sd	Sudan	114
tn	Tunisia	623
ye	Yemen	328
Total		**4,001**

Table B.1: Distribution of evaluation text samples across target dialects.

Dialect	Duration	# words (AR/EN)
Algeria	0:55:03	8,032/10,559
Egypt	0:59:12	10,116/12,228
Jordan	0:58:54	8,804/12,188
Mauritania	0:56:15	10,116/12,228
Morocco	1:00:23	11,959/15,796
Palestine	0:58:22	8,833/11,729
UAE	0:55:59	8,766/12,163
Yemen	1:02:30	9,359/13,296
Total	**7:46:38**	**75,985/ 100,187**

Table B.2: Duration (h:mm:ss) and number of words per dialect of the NADI 2026 official speech translation evaluation set. # words represent the number of words per dialect in the original dialectal transcription (AR) and the corresponding English translation (EN).

to better contextualize the submission we received.

Task 4 Spoken Language Translation: We use `whisper-large-v3` as a baseline model to generate the English translation of all the considered Arabic dialects. Whisper is an end-to-end multi-task speech model that adopts a transformer-like encoder-decoder architecture. It natively handles language identification, speech transcription, and speech translation directly into English text.

Task 5 Spoken Language Understanding: We use `whisper-small` as the backbone for both of Task 5's baselines. For the intent recognition subtask, we simply add masked average pooling on the encoder output and pass this to a linear layer for classification. For slot filling, we directly fine-tune `whisper-small` to add the slot labeling as part of the output transcription. In both parts we use SLURP-TN for training data.

D Additional Results

Due to space constraints we provide tables of results for the remaining subtasks:

- Subtask 1.3 Code-switched ASR: Table D.1.
- Subtask 2 SDID: Table D.2.
- Subtask 3 TTS: Table D.3.
- Subtask 4 SLT: Table D.4
- Subtask 5.1 Intent Classification: Table D.5
- Subtask 5.2 Slot Fillings: Table D.6

E Complete Submission Descriptions

E.1 Subtask 1.1: Robust Dialectal ASR

Wifaq (Rank 1) (Alkurbi, 2026). Wifaq fine-tuned four character-level connectionist temporal classification (CTC) recognizers (Graves et al., 2006) from `w2v-bert-2.0` using different training stages and mixtures of NADI, Casablanca, selected Omnilingual ASR subsets (Omnilingual ASR et al.,

	Official		Hamza-norm.						
Team	**WER↓**	**CER↓**	**WER↓**	**CER↓**	**Backbone**	**Adaptation**	**Data+**	**CS-targeted**	**Ensemble**
Abjad AI	**14.41**	5.55	**14.12**	5.43	Cohere	Full FT	✓	✓	✓(5)
Itgan	14.49	**5.38**	14.22	**5.27**	`whisper-large-v3`	LoRA			✓(3)
Thakaa	14.53	6.34	14.26	6.26	Heterogeneous	Full FT	✓		✓(10)
KAND CA	15.21	5.86	14.91	5.73	`whisper-large-v3`∗	Full FT	✓	✓	
NAMAA	16.98	6.58	16.68	6.46	`FarukSTT` (Whisper)	LoRA		✓	
Ar_India	73.76	49.56	71.81	48.91					
Baseline	88.60	66.14	87.91	65.61	`whisper-large-v3`				

Table D.1: Official Subtask 1.3 (Code-switched ASR) results and reported system characteristics for the six teams that provided system descriptions, ordered by percentage WER; percentage CER and the hamza-normalized variants of both metrics are also shown, and lower is better. *Cohere* refers to the `cohere-transcribe-arabic-07-2026` model. ∗ Finetuned dialectal checkpoint. **Adaptation**: distinguishes full fine-tuning from LoRA; **Data+**: whether additional data was used on top of the Subtask 1.3 releases, excluding checkpoint pretraining; **CS-targeted**: marks measures aimed specifically at code-switched segments, such as adding non-Arabic speech, oversampling Latin-script references, or lifting Arabic-only decoding constraints; and **Ensemble**: indicates the number of components used in the final ensemble. All teams except THAKAA reported acoustic augmentation and used an external decoding language model, and KAND CA alone applied output post-processing. AR_INDIA did not submit a system paper.

	Accuracy ↑	$\mathbf{C}_{avg}$ **↓**	**Audio**	**Text Branch**	**Ensemble**	**Augm.**	**OOD**
Thakaa	**56.15**	11.31	Ara-BEST-RQ	✓	✓(5)		
Salesteq	54.67	**8.64**					
Lynx	53.53	12.47	Cohere	✓			✓
AI Elites	52.51	11.39	Cohere			✓	✓
KAND CA	51.14	13.26	Cohere	✓	✓(2)		
Abjad Ai	41.57	14.73	Cohere			✓	
Resonate	40.89	16.78	WavLM			✓	
FARABI-Fraunhofer-IAIS	39.18	16.95					
CodeZone Research group	39.18	14.69	SB+Whisper		✓(2)		
Namaa	37.70	16.74	SB+Whisper		✓(4)		
Ar_India	31.21	20.26					
ArabSpeech Minds	30.64	16.97					
Baseline	28.70	20.02	SB				

Table D.2: Results from Subtask 2 (Spoken Dialect Identification) alongside submission approach. Best results in bold. We report $C_{avg} * 100$ for readability. **Audio**: audio models used with shorthands for *Cohere* (`cohere-transcribe-arabic-07-2026`) *SB* (Speech Brain `lang-id-voxlingua107-ecapa`), and *Whisper* for any of the Whisper family models; **Text Branch**: if the approach uses a multimodal text+speech pipeline; **Ensemble**: number of ensemble component models; **Augm.**: if any data augmentation approach was used; **OOD**: if teams applied specific methods to enhance out-of-domain performance. SALESTEQ, FARABI-FRAUNHOFER, AR_INDIA, and ARABSPEECH MINDS did not submit system papers.

2025), and MoulSot Moroccan Arabic (Atlasia, 2026). Each model used beam search with a five-gram KenLM trained on speech transcripts and approximately 440,000 dialectal tweets; word-level ROVER combined their hypotheses.

Resonate (Rank 2) (Shoaib et al., 2026). Resonate fully fine-tuned the Arabic-specialized `cohere-transcribe-arabic-07-2026` model on official data expanded through neural denoising (Zhao et al., 2025), k-nearest-neighbor voice conversion (Baas et al., 2023), and using a low-pass filter to better match audio energy characteristics. It conditioned the decoder on the country label and used duration-aware SpecAugment for short clips (Park et al., 2019). Beam-search output underwent dialect-specific orthographic canonicalization.

Thakaa (Rank 3) (Otaif et al., 2026). Thakaa selected ROVER committees from seven candidates: one Cohere model, four NeMo FastConformer (Rekesh et al., 2023) variants, one Whisper LoRA model, and one Qwen3-ASR model (Yang et al., 2025a). Its strongest component was trained on Subtasks 1.1 and 1.3, while other candidates used KenLM shallow fusion or LoRA adaptation. Validation-based search selected one committee for five dialects and specialized committees for Algeria, Mauritania, and Yemen.

Dialect / Country	UTMOS ↑		NISQA-MOS ↑		NISQA-TTS ↑		WER ↓		CER ↓	
	K	B	K	B	K	B	K	B	K	B
United Arab Emirates	2.73	**3.10**	3.75	**4.29**	3.45	**3.85**	**31.31**	31.71	15.45	**8.45**
Algeria	2.60	**3.13**	3.81	**4.31**	2.96	**3.95**	44.63	**40.82**	13.01	**10.13**
Egypt	2.81	**3.15**	2.75	**4.30**	3.75	**3.94**	**20.49**	28.59	**5.37**	7.33
Jordan	2.60	**3.10**	4.21	**4.29**	3.18	**3.91**	32.84	**30.03**	8.19	**6.98**
Morocco	**3.23**	3.13	3.52	**4.39**	3.05	**3.87**	55.46	**27.34**	21.62	**6.17**
Palestine	2.58	**3.10**	4.23	**4.38**	3.09	**3.90**	**20.16**	20.33	5.66	**5.33**
Saudi Arabia	2.79	**3.09**	3.85	**4.28**	3.45	**3.84**	14.28	**13.59**	4.87	**3.57**
Sudan	**3.19**	3.10	**4.46**	4.21	**3.98**	3.96	35.71	**29.98**	9.96	**7.52**
Tunisia	3.04	**3.13**	2.74	**4.30**	3.20	**3.94**	44.52	**34.18**	12.23	**7.75**
Yemen	2.17	**3.09**	2.65	**4.29**	2.46	**3.87**	47.92	**31.31**	25.12	**7.97**
Average	2.72	**3.11**	3.57	**4.31**	3.21	**3.91**	32.48	**29.28**	11.40	**7.22**

Table D.3: Detailed dialect-level evaluation of the KAND CA submission (column **K**) for Task 3 (TTS), against the XTTS-v2 baseline (column **B**). UTMOS, NISQA-MOS, and NISQA-TTS evaluate speech quality, with higher scores indicating better performance, while WER and CER (percentages) measure transcription errors, with lower values indicating better intelligibility. Bold values indicate the best performance.

	Overall		Dialect BLEU							
Team	**BLEU ↑**	**chrF ↑**	**ALG**	**EGY**	**JOR**	**MAU**	**MOR**	**PAL**	**UAE**	**YEM**
SENSE	**14.19**	**34.11**	**12.16**	**19.84**	**27.46**	**6.84**	**8.51**	**20.32**	**0.39**	**16.07**
ABJAD AI†	5.77	22.94	7.08	0.14	20.65	3.53	0.21	0.39	0.18	10.97
Baseline	5.52	20.28	7.95	0.06	21.6	3.22	0.07	0.35	0.07	11.0

Table D.4: Evaluation results for Task 4 (Spoken Language Translation). Overall performance is reported using BLEU and chrF, while dialect-level performance is reported using BLEU; higher scores indicate better performance. Dialect abbreviations are ALG (Algeria), EGY (Egypt), JOR (Jordan), MAU (Mauritania), MOR (Morocco), PAL (Palestine), UAE (United Arab Emirates), and YEM (Yemen). Bold values indicate best performance. † This submission is a post-evaluation phase run which aimed to minimize an unintended use of development data in system training.

Rank	Team	F1 ↑	Accuracy ↑
1	**Lynx**	**76.86**	**75.73**
2	CIS-OpenSLU	71.54	72.50
3	A3da2 Messi	69.92	70.07
4	Aslema	66.93	66.13
5	FARABI-Fraunhofer	64.69	65.93
6	Baseline	11.08	13.35

Table D.5: Official results for Task 5.1 (Intent Recognition). Systems are ranked by weighted F1; higher weighted F1 and accuracy indicate better performance.

Rank	Team	CoER ↓	CVER ↓
1	**Aslema**	**59.53**	**94.20**
2	Zila	72.78	100.36
3	Baseline	100.36	120.95

Table D.6: Official results for Task 5.2 (Slot Filling). Systems are ranked by Concept Error Rate (CoER); lower CoER and Concept-Value Error Rate (CVER) indicate better performance.

KAND CA (Rank 4) (Wasfy and Anber, 2026). The team continued full-model adaptation of `whisper-large-v3` on NADI and Casablanca data with replay; its broader Arabic training mixture included Lahgtna (KAND CA, 2026), Common Voice (Ardila et al., 2020), FLEURS (Conneau et al., 2023), MASC (Al-Fetyani et al., 2022), and additional Algerian speech. Inference used overlapping chunks with repetition removal. Eight dialect-specific mT5-large spelling correctors were trained on synthetic corruptions of the available official transcripts and applied to every hypothesis to better match the task's orthographic conventions.

Ar_India (Rank 5) . The Ar_India team did not submit a system paper.

Nile (Rank 6) (Alshrief et al., 2026). Nile used a two-stage pipeline based on `whisper-large-v3-turbo`; It first adapted the model on an unspecified 200,000-recording corpus augmented with noise, reverberation, gain changes, and background music, and then fine-tuned the

checkpoint on the official competition training set. This separated noise-robustness training from task-specific dialect and transcription adaptation.

Itgan (Rank 7) (Almajai, 2026). Itgan trained rank-32 LoRA adapters using only the official data. A pooled `whisper-large-v3-turbo` adapter was specialized for Algeria, Egypt, Mauritania, and Morocco, while validation favored a non-Turbo Whisper backbone for the other four countries. Country labels routed each utterance, and greedy Arabic decoding included a hypothesis-length guard.

Sabaa (Rank 8) (Sabaa and Sabaa, 2026). Sabaa trained a separate rank-32 `whisper-large-v3-turbo` LoRA adapter for each dialect using only official data; this outperformed a pooled adapter. Beam-search decoding used beam size 8 and a negative length penalty, followed by conservative dialect-specific spelling normalization.

Salesteq (Rank 9) . The Salesteq team did not provide a system paper.

E.2 Subtask 1.2: Mixed dialect ASR

Salesteq (Rank 1) The Salesteq team did not submit a system paper.

Namaa Community (Rank 2) (Abdelazi et al., 2026). They used `cohere-transcribe-arabic-07-2026` in a zero-shot setting, without task-specific fine-tuning. The Task 1.2 development set was used only for system selection, with no additional ASR training data included in the final configuration. Several alternatives were explored, including quantized inference, different decoding settings, orthographic normalization, and consensus-based hypothesis combination. The final system used full-precision inference without ensembling or additional text post-processing.

Itgan (Rank 3) (Almajai, 2026). The submission was based on `whisper-large-v3` and `whisper-large-v3-turbo`, with LoRA used for parameter-efficient adaptation. Different model variants were trained using distinct data configurations, and one component additionally incorporated 12.8K utterances from Task 1.1. No data external to NADI 2026 were used. The final system combined five recognizers using ROVER: three adapted `whisper-large-v3` variants, one adapted `whisper-large-v3-turbo` model, and one zero-shot `whisper-large-v3` model. Inference used the Faster-Whisper [10] interface for Whisper with beam size 5 and Silero VAD (Silero Team, 2024), with previous-text conditioning disabled.

KAND CA (Rank 4) (Wasfy and Anber, 2026). They used `whisper-large-v3` with sequential full-model fine-tuning from a checkpoint previously adapted to Arabic dialectal speech. During Task 1.2 adaptation, 1.5K Task 1.1 utterances and 1K Casablanca clips were replayed to preserve previously learned dialectal information, while earlier adaptation stages also incorporated additional Arabic dialect resources. The final system used a single model with chunked long-form decoding followed by repetition removal.

E.3 Subtask 1.3: Code-switched Tunisian ASR

Abjad AI (Rank 1) (Alhumid et al., 2026). Abjad AI fully fine-tuned `cohere-transcribe-arabic-07-2026`, keeping FP32 master weights under bfloat16 autocasting because pure bfloat16 rounded small AdamW updates to zero, and masking prompt positions from the loss. The official data was supplemented with text-to-speech-synthesized Tunisian speech, nine hours each of French and English read speech, and ESC-50 noise (Piczak, 2015), with synthetic recordings augmented more strongly than real ones; both online and offline waveform augmentation were used. The submission uniformly averaged five checkpoints trained under complementary data and augmentation configurations, averaging floating-point parameters in FP32 and retaining integer BatchNorm counters from one reference checkpoint.

Itgan (Rank 2) (Almajai, 2026). Itgan used only the shared-task corpus and trained rank-32 LoRA adapters on all attention and feed-forward projections of `whisper-large-v3`, 57.7 M parameters or 3.6% of the model, against targets processed by the organizer's cleaned-transcription form so that training matched exactly the text the metric scores. After a first run memorized heavily, raising LoRA dropout to 0.15 and enabling SpecAugment shrank the generalization gap by a factor of 3.5. Two combination stages followed with no further training: four adapters from two independent runs were av-

[10] https://github.com/SYSTRAN/faster-whisper

eraged in weight space by concatenation, which yields the exact mean of the adapter updates as an ordinary higher-rank adapter, and then that model combination was used to pivot a three-system word-level ROVER vote that modified 63 of the 2,123 blind utterances. Doubling the LoRA rank, collapsing hamza variants and centroid hypothesis selection all failed to help.

Thakaa (Rank 3) (Otaif et al., 2026). Thakaa submitted a ten-system word-level ROVER ensemble spanning CTC encoders (`w2v-BERT-2.0`, `mms-1b`), an encoder-decoder recognizer (`cohere-transcribe-arabic-07-2026`), an audio-LLM (`Qwen3-ASR-1.7B`) and two `whisper-large-v3` fine-tunes. Training added roughly 6,900 fully labeled TUNIFRA and Tunisian SLURP-derived utterances with no pseudo-labeling. A 4-gram KenLM gave shallow-fusion CTC decoding about 20% relative WER over greedy decoding, while the autoregressive systems used beams of 5 to 10 with repetition constraints. Committee composition was searched over all checkpoints with bootstrap significance testing anchored on the strongest CTC system.

KAND CA (Rank 4) (Wasfy and Anber, 2026). KAND CA began from a broadly dialect-adapted `whisper-large-v3` checkpoint, which it measured as worth about 4.3 WER points over an English base after identical fine-tuning. Full fine-tuning with SpecAugment was followed by a single fresh two-epoch schedule over 70,595 clips combining roughly 46,000 external gold Tunisian-French clips with TEDx replay, reported as the decisive step; warm continuation and further epochs both regressed. Decoding was greedy, as beam search and n-gram blocking hurt, and post-processing was restricted to an accent-only filter that accepted a French spellchecker fix only when unaccenting it reproduced the original token, touching five tokens. Every learned corrector degraded results, most severely a `Qwen3.5-4B` LoRA corrector at 5.9 points.

NAMAA Community (Rank 5) (Abdelazi et al., 2026). NAMAA Community adapted FarukSTT [11], a Tunisian Whisper derivative, in two LoRA stages while using the `whisper-large-v3` processor so Arabic, French and English tokens could co-occur. Stage 1 trained rank-32 adapters on the query and value projections and merged them; stage 2 started from that merge, added the publicly released legacy test split, and duplicated every Latin-script example to increase code-switch exposure, giving 38,106 examples from 25,730 sources. Adapters covered the attention projections, feed-forward layers and output projection, with checkpoint selection on validation WER. Inference used beam size 3 with forced decoder IDs and token suppression disabled to lift the Arabic-script restriction, and raw transcriptions were submitted because the scorer normalizes both sides.

[11] https://huggingface.co/medyas/FarukSTT

Ar_India (Rank 6) The Ar_India team did not submit a system paper

E.4 Subtask 2: Spoken Dialect ID

Thakaa (Rank 1) (Otaif et al., 2026). This multimodal system use Ara-BEST-RQ (Elleuch et al., 2026b) for its audio component alongside multiple text pipelines. Two text pipelines were trained, one using standard transcriptions, and the second stream using fused transcriptions, auxiliary branch loss and transcription-view dropout. Each of the pipelines consisted of a Cohere Arabic transcription branch and a `whisper-large-v3` branch. Training consisted of a mix of open resources totally 265 hours from eight distinct ADI and ASR datasets.

Salesteq (Rank 2) The Salesteq team did not submit a system paper.

Lynx (Rank 3) (Atou et al., 2026). Team Lynx uses `cohere-transcribe-arabic-07-2026` as a multimodal backbone with a text-branch utilizing TF-IDF features of the transcripts directly, while the acoustic branch uses the Cohere model as a frozen feature extractor before classifying the utterances with a two layer LSTM model. Their pooling method for the acoustic branch is of note, and involves temporal mean pooling first before passing all layers as a sequence to their LSTM classifier.

AI Elites (Rank 4) (Iskandar et al., 2026). The AI Elites team focused on an speech-only model, taking `cohere-transcribe-arabic-07-2026` and finetuning it for SDID on the micro ADI20 dataset. Modifications to the model include using a learnable weighted sum through all encoder layers. Data augmentation was comprehensive and included noise and codec compression; speech and gain permutation; as well as SpecAugment (Park et al., 2019) and random cropping. Key to their approach is using out-of-domain dataset

for checkpoint selection, of which they used both Casablanca (Talafha et al., 2024) and the MADIS-5 (Abdullah et al., 2025) as a secondary check.

KAND CA (Rank 5) (Wasfy and Anber, 2026). This multimodal submission used an ensemble, the first part of which was a finetuned FastConformer using an attention pooling head on top of `cohere-transcribe-arabic-07-2026`. To this they included the same classifier applied to a LavaSR [12] speech-enhanced version of the target audio, weighted evenly with the non-enhanced audio. Finally they included a text route classifier weighted at 30%, which consisted of a finetuned `whisper-large-v3` model trained on the 1.1 subtask, fed to a BERT based dialect classifier.

Abjad Ai (Rank 6) (Alhumid et al., 2026). This team also used `cohere-transcribe-arabic-07-2026` as starting point, and finetuned on the micro ADI20 dataset using a simple two layer classifier head with GELU activation and dropout on top of the fully frozen conformer model. For data augmentation, they applied a mix of methods including gain, additive noise, speed perturbation, and time-shifting. To reduce channel-sensitivity normalize loudness as well as apply a simulated bandwidth reduction.

Resonate (Rank 7) (Shoaib et al., 2026). This team uses `wavlm-base-plus` (Chen et al., 2022) as a base model with output passed to context-aware multi-head factorized attentive pooling to a final linear layer for classification. Models were trained with additive-angle margin softmax. A 100hr per dialect subset of ADI17 was used alongside the new additions from ADI20. Data filteration included removing short clips or clips with little speech as determined by a VAD. All training utterances were cut down to five seconds for training, meaning longer samples could lead to more distinct clips for training. The starting point of these samples were altered every few epochs. Data augmentation included speed, reverb, additive noise, frequence and time dropping.

FARABI-Fraunhofer (Rank 8) The FARABI-Fraunhofer team did not submit a system paper.

CodeZone Research group (Rank 9) (Yassine and Bichi, 2026). This group uses an ensemble of speech models including a pretrained VoxLingua107 ECAPA-TDNN system [13] and the pretrained ADI20 Whisper model from Elyadata [14]. Both models are finetuned on the ADI20-10hr using a mix of data augmentation methods.

Namaa Community (Rank 10) (Abdelazi et al., 2026). This four component ensemble approach focused on models trained on the ADI20 dataset. The base models used included a VoxLingua107 pretrained ECAPA-TDNN system and finetuned with on the ADI20-10hr split; `whisper-large-v3` finetuned with LoRA on the ADI20-10hr split; `whisper-large-v3` trained on the ADI20-53hr split; and finally a `whisper-large-v3` model finetuned on the ADI17 dataset. Logits were combined using a weighted sum approach picked using the development set.

Ar_India (Rank 11) The Ar_India team did not submit a system paper.

Arabspeech Minds (Rank 12) The Arabspeech Minds team did not submit a system paper.

E.5 Task 4 SLT

Sense (Rank 1) Team Sense did not submit a system paper.

Abjad AI (Rank 2) (Alhumid et al., 2026). The system used `whisper-large-v3` with full-model fine-tuning for Arabic-to-English speech translation. Two adaptation strategies were explored: intermediate fine-tuning on an internal out-of-domain Arabic speech corpus followed by task-specific adaptation, and direct fine-tuning on the shared-task data. The latter yielded better development performance and was adopted for the final system. To address the limited amount of training data, the speech was augmented using Gaussian noise, time stretching, pitch shifting, and temporal shifting, and the training set was upsampled to approximately 2.5 hours. The final model was fine-tuned directly on the augmented and upsampled Task 4 data. During a review of the system, the team discovered some of their training data used was shown to overlap with the development dataset, the reported results are thus a re-run of their system with this overlap minimized.

[12] https://github.com/ysharma3501/LavaSR

[13] https://huggingface.co/speechbrain/lang-id-voxlingua107-ecapa

[14] https://huggingface.co/Elyadata/ADI-whisper-ADI20

E.6 Task 5.1 Intent Classification

Lynx (Rank 1) (Atou et al., 2026). Team Lynx adopted a cascaded system utilizing `cohere-transcribe-arabic-07-2026` as a frontend to transcribe the utterances before applying extracting word and character-level n-gram features for TF-IDF representations. Character n-grams used n$\in \{2, 5\}$, while they only used word unigrams and bigrams. These features are fused and provided to a one-rest logistic regression classifiers trained for each of the intents. To capture unknown intent scenarios, they also employ `MARBERTv2` (Abdul-Mageed et al., 2021) model trained to for intent classification on the MASSIVE corpus (FitzGerald et al., 2023), utterances that fall below a threshold probability for sum of the corresponding categories of the 5.1 subtask are labeled as unknown.

CIS-OpenSLU (Rank 2) (Khaled, 2026). The system followed a two-stage pipeline in which `whisper-large-v3-turbo`, implemented with Faster-Whisper, was first used to transcribe the Tunisian Arabic speech. Intent recognition was then performed using a linear SVM trained on word- and character-level TF-IDF features extracted from the ASR transcripts. The official training and validation sets were combined, resulting in 3,272 labeled utterances, without external labeled data or synthetic augmentation. To handle intents not represented in the training data, confidence-based open-set recognition was applied using the maximum SVM decision score, with low-confidence predictions assigned to an *unknown* class.

A3da2 Messi (Rank 3) The A3da2 Messi team did not submit a system paper.

Aslema (Rank 4) (Shafiq et al., 2026). The system used `Qwen3-Omni-30B-A3B-Instruct` with joint LoRA fine-tuning for intent recognition and slot filling. Model selection was performed after evaluating several multimodal foundation models in both zero-shot and fine-tuned settings. The official training set, containing approximately 2.7K examples, was augmented with synthetic Tunisian Arabic data generated using Gemini and filtered with an LLM-based quality check. Corresponding speech was synthesized using VoxCPM and further filtered for acoustic quality, yielding approximately 23K additional speech–text pairs. The final model was trained on the combined original and synthetic data.

FARABI-Fraunhofer (Rank 5) The FARABI-Fraunhofer team did not submit a system paper.

E.7 Task 5.2 Slot Filling

Aslema (Rank 1) (Shafiq et al., 2026). The system used `Qwen3-Omni-30B-A3B-Instruct` and jointly modeled intent recognition and slot filling through LoRA-based fine-tuning. Training combined the official dataset, containing approximately 2.7K examples, with a substantially larger synthetic set. Synthetic Tunisian Arabic utterances with intent and slot annotations were generated using Gemini, filtered for quality, and converted to speech using VoxCPM (Zhou et al., 2026), resulting in approximately 23K additional speech–text pairs. The final model was fine-tuned on the combined original and synthetic data, enabling direct prediction of slot labels and their corresponding values from speech.

Zila (Rank 2) The Zila team did not submit a system paper.